



Scuola Superiore
Sant'Anna

Master of Science in Bionics Engineering

Academic Year 2025-2026

Pontedera, April 16, 2026

Bioinspired and Soft Robotics

Soft Robotics Technologies

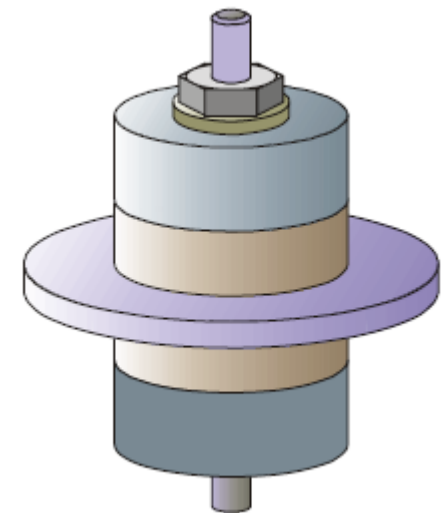
PiezoElectric actuators

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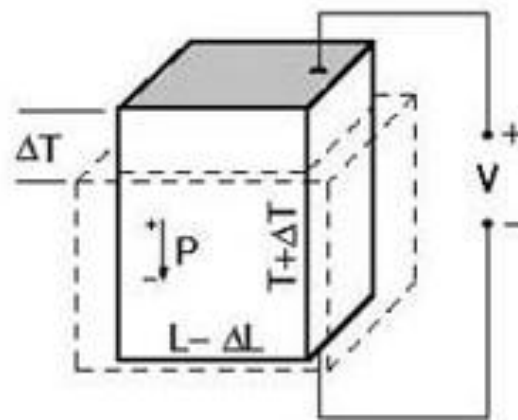
History

- Piezoelectricity was first observed in some natural materials (quartz, tourmaline and Rochelle salt), which exhibited electrical polarization as the result of an applied mechanical load by Pierre and Jacques Curie in 1880.
- Lippmann predicts the inverse piezoelectric effect in 1881, and the Curies verify this experimentally.
- The first engineering application of piezoelectricity was devised by Langevin, who laid the foundations of ultrasonic submarine detection in 1916 (quartz-steel sandwich transducer).



Definition

The **piezoelectric effect** can be described as modification of the polarization of a dielectric **arising from the mechanical energy of the stress**. Materials exhibiting such an effect are said to be piezoelectric materials. The piezoelectric effect is reversible in the sense that when an electric field is applied, a mechanical strain will arise. This is the inverse piezoelectric effect.



Definition

In what are called dielectric materials, **the application of an external electric field leads to electric dipoles**. This phenomenon is known as **electric polarization** and is usually characterized by the vector magnitude **P**. In these materials, the electrical charge per unit area is known as **electric displacement** and is denoted by the vector magnitude **D**. The electrical state of a dielectric is then determined according to the following expression:

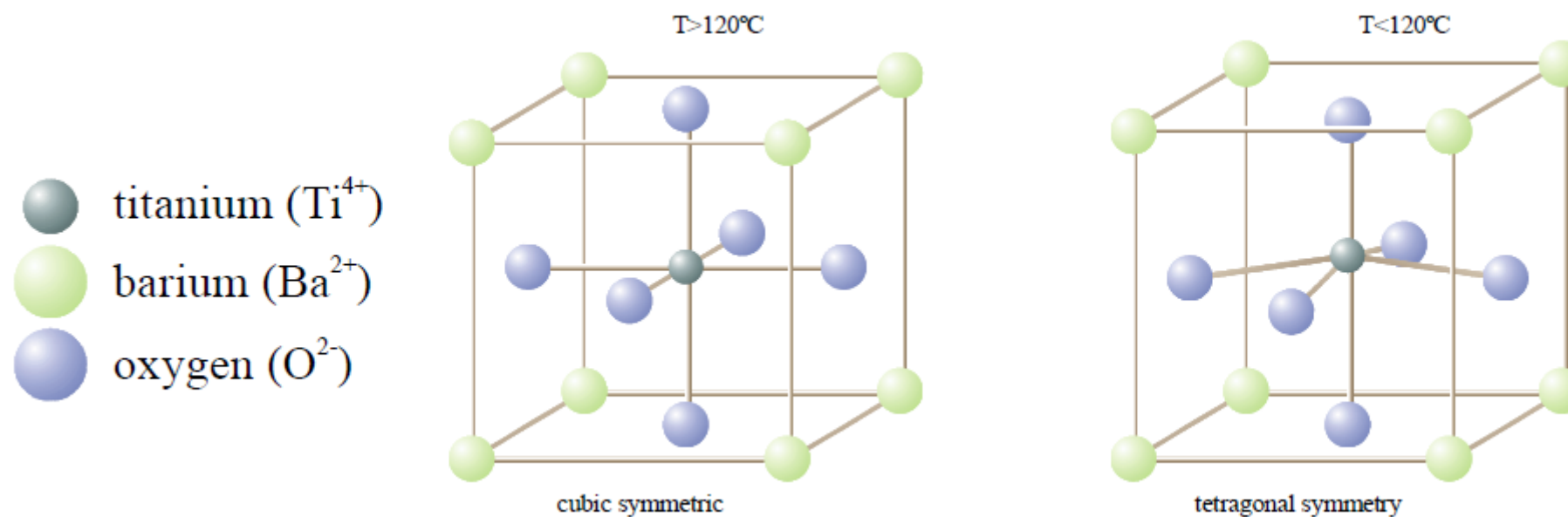
$$\begin{aligned}\vec{\mathbf{D}} &= \epsilon_0 \vec{\mathbf{E}} + \vec{\mathbf{P}} \\ &= \epsilon \epsilon_0 \vec{\mathbf{E}}\end{aligned}$$



Definition

Ferroelectricity

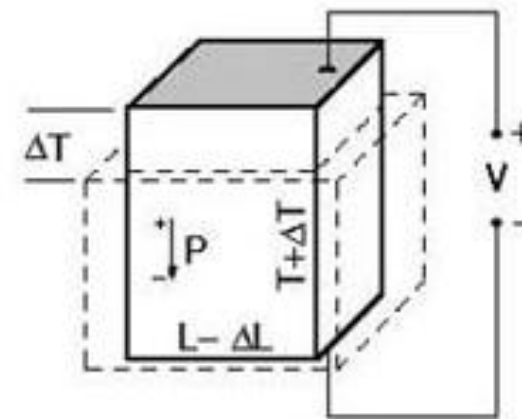
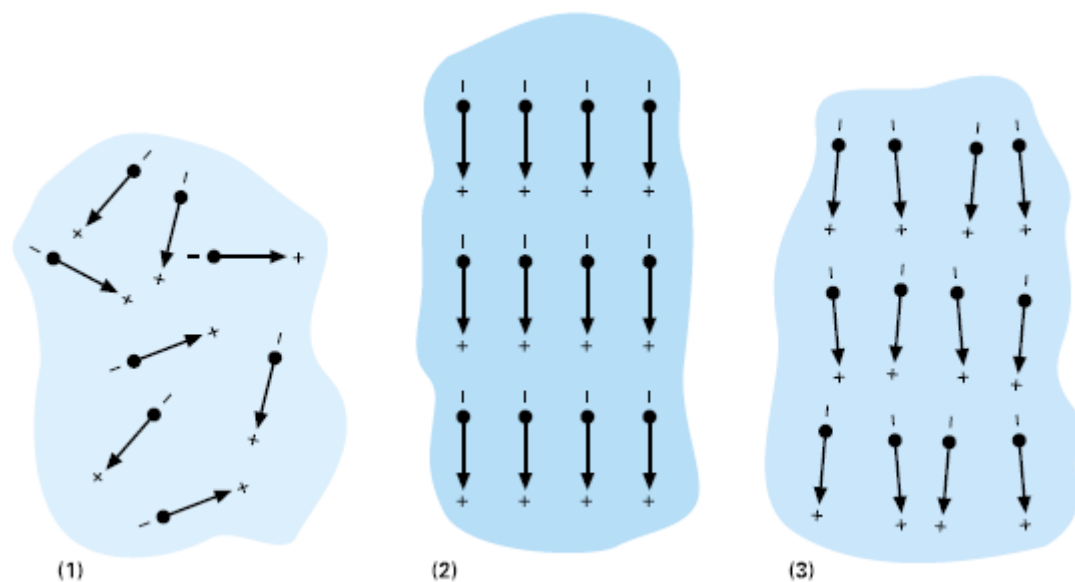
- Above 120°C (Curie point), barium titanate has a face-centered cubic (fcc) symmetry and there is no net polarization of charge
- Below 120°C, the structure changes to tetragonal, shifting the positions of the ions and causing a concentration of positive and negative charges towards opposite ends of the crystal



Definition

When the crystal structure of the dielectric material is such that in the absence of external electric fields the centers of positive and negative electrical charges do not coincide, the material is said to exhibit **spontaneous polarization**. If this polarization can be modified when an external electric field is applied, the material is said to be **ferroelectric**. Ferroelectric materials are not generally piezoelectric. This is mainly because they are isotropic materials; that is, there is no dominant direction for polarization.

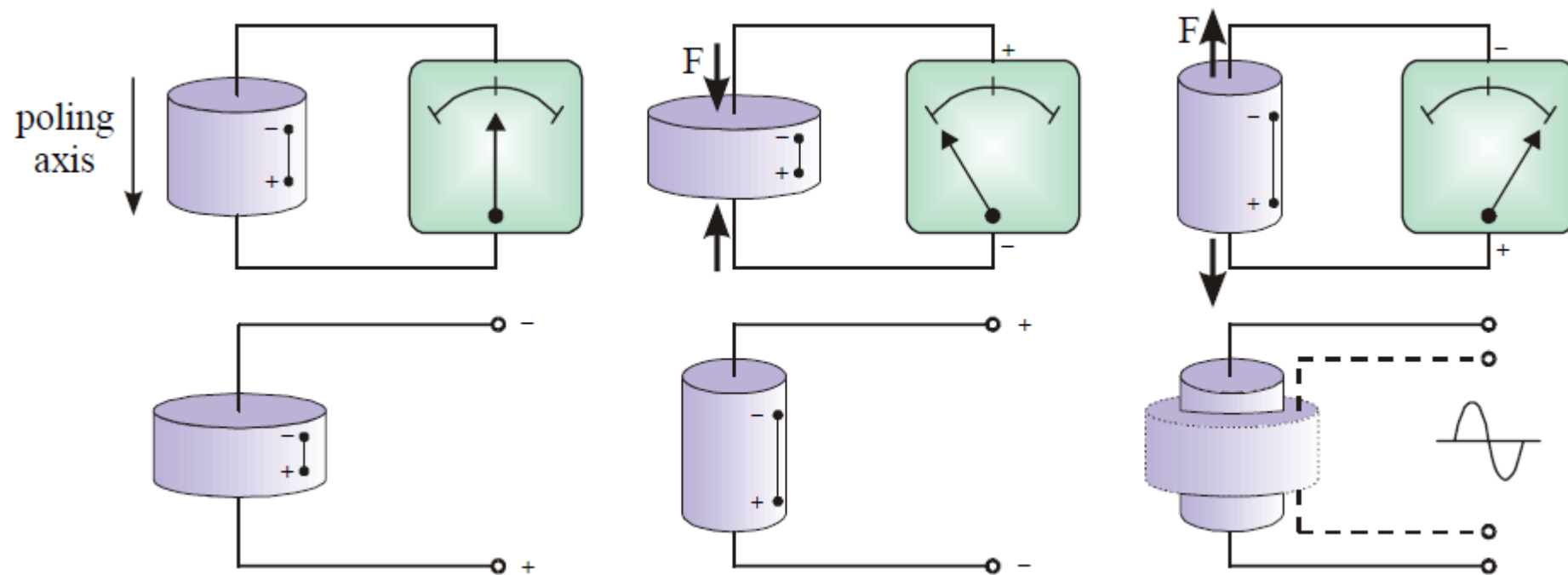
In order to serve as piezoelectric materials, ferroelectric materials must be subject to an external electric field, the poling field. Therefore, it is the poling process that is responsible for the appearance of piezoelectric properties in ferroelectric materials.



Definition

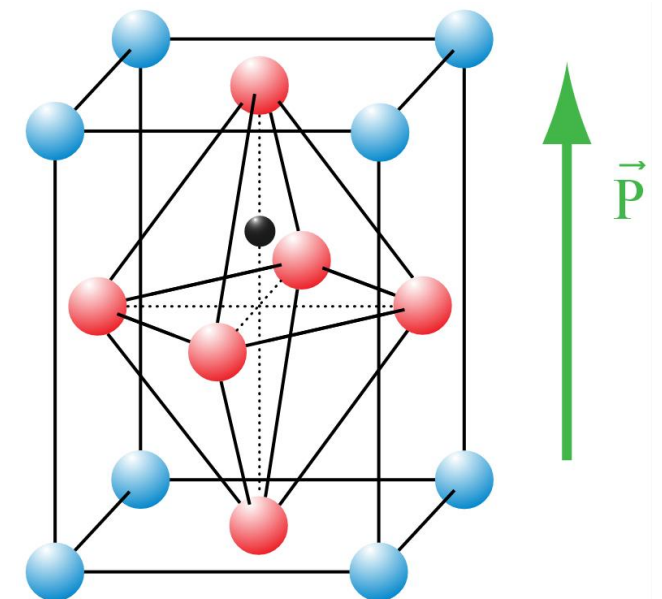
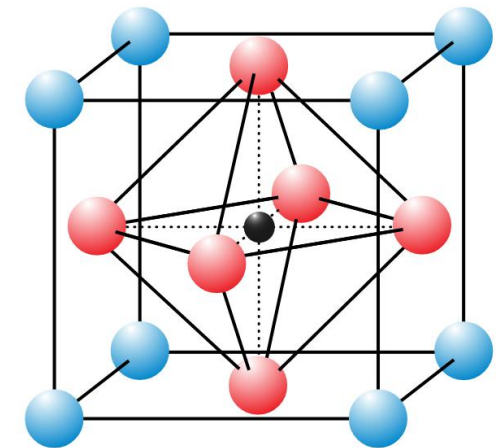
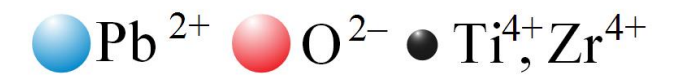
Piezoelectricity: a dual effect

- The direct piezoelectric effect is the ability of materials to produce electric charges when subjected to a mechanical deformation
- The converse piezoelectric effect is the ability of the same materials to deform when placed under an electric field



Materials

Structure	Material
Single Crystals	Quartz Lead Magnesium Niobate (PMN-PT and PZN-PT)
Ceramics	Lead Zirconate Titanate (PZT) Lead Metaniobate (LMN) Lead Titanate (LT) Lead Magnesium Niobate (PMN)
Polymers	Polyvinylenedifluoride (PVDF)
Composites	Ceramic-polymer Ceramic-glass



Piezoelectric materials as actuators

PROs and CONs

PRO

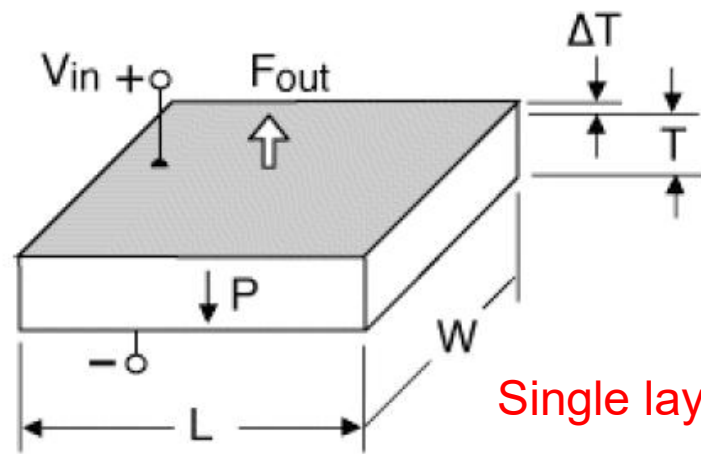
- High torque
- Fast
- No magnetic fields
- Low inertia
- Strong in compression

CON

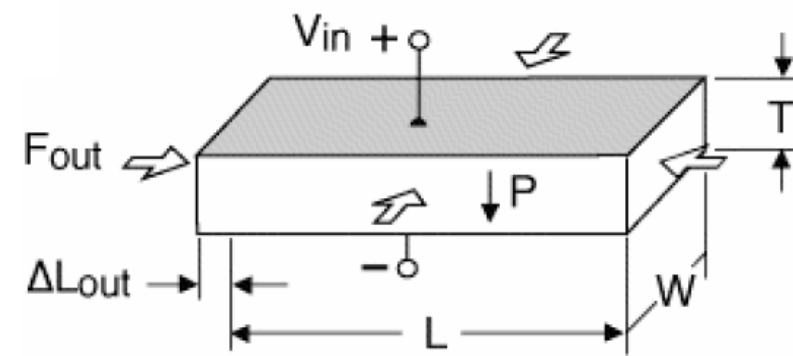
- Small deformation (0.1% to 0.18%)
- Relatively low stability (degrade gradually depending on composition, manufacturing process)
- Electrical, mechanical and thermal limitations
- Weak in tension
- Fatigue



Deformation modes



Single layer d33

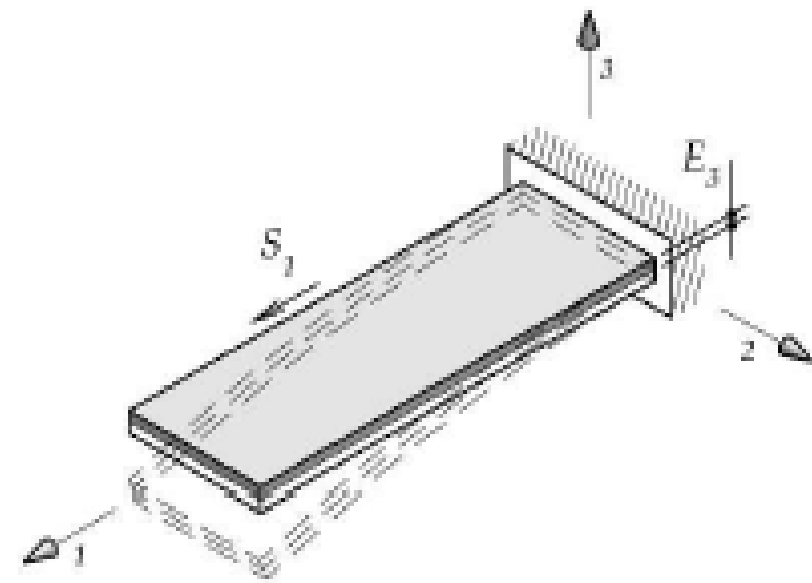
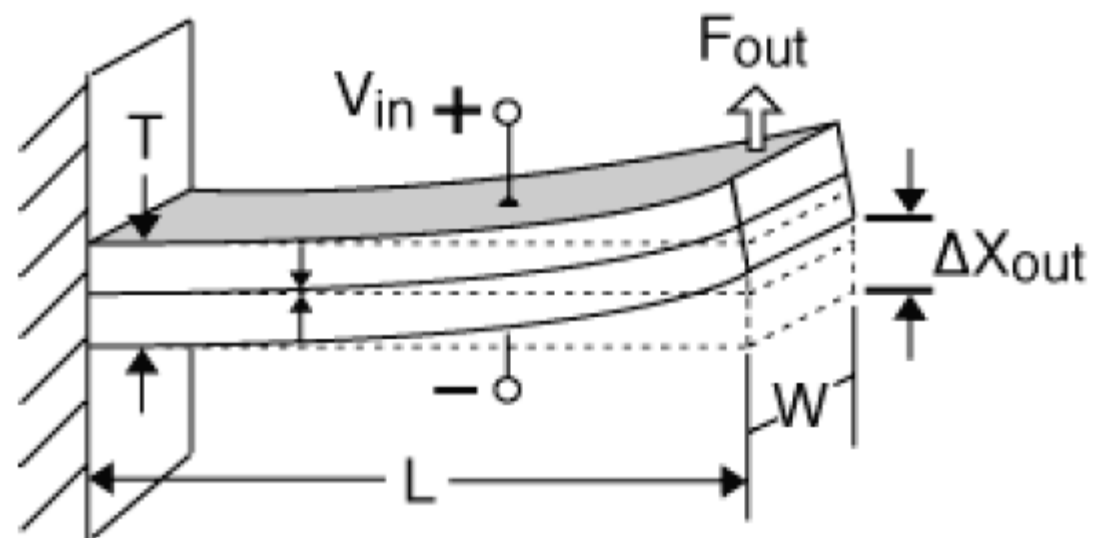


Transverse excitation d31



Bending

Mono- or Multi-layer laminates bending when an electric field is applied



Bending

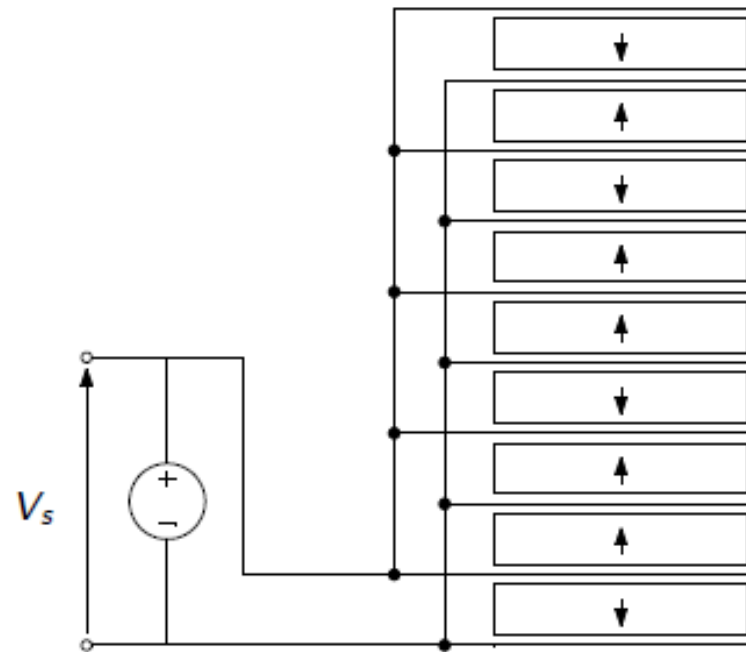
Example:

<https://www.piceramic.com/en/products/piezoceramic-actuators/bender-actuators/pl112-pl140-picma-bender-103000/>



Stack

Mono- or Multi-layer laminates contracting when an electric field is applied



$$C_0^j = \epsilon^S \frac{A}{t} \quad \text{Single layer capacity}$$

$$C_0 = \sum_{j=1}^L C_0^j = L \epsilon^S \frac{A}{t} \quad \text{Total capacity}$$

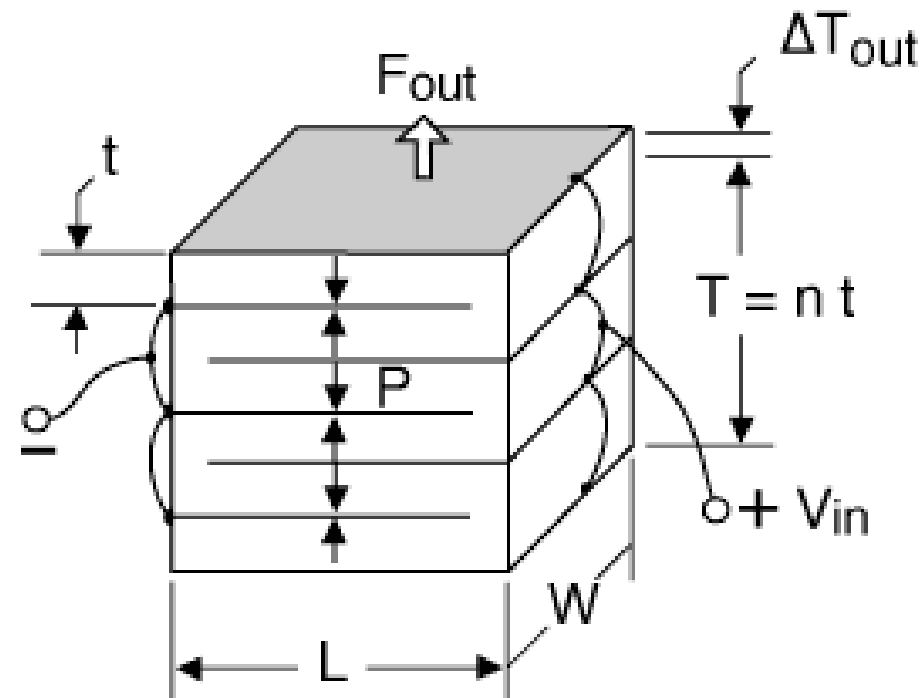
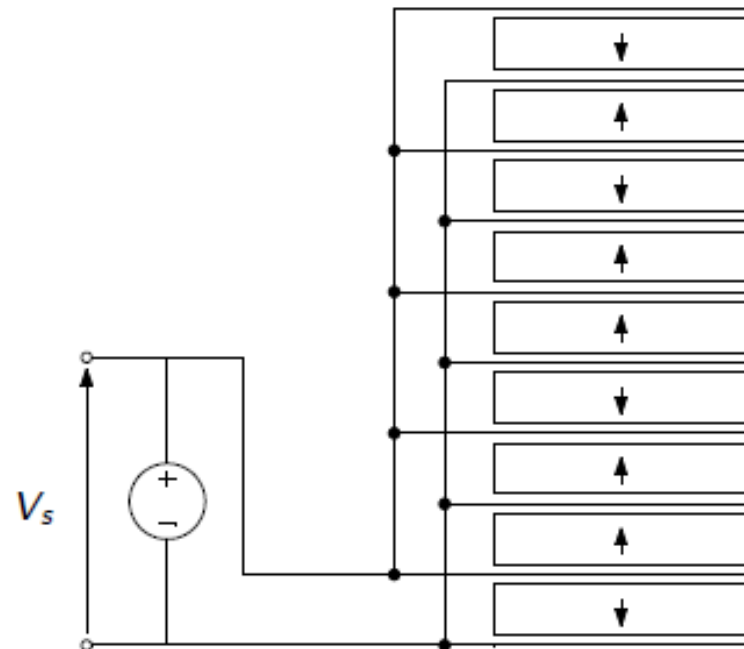


Stack

Mono- or Multi-layer laminates contracting when an electric field is applied

- Meccanically in series
- Electrically in parallel

Given the same voltage,
the strain is multiplied by
the number of layers



Single layer d_{33}

$$E = V/d$$

$$d = T/n = t$$

$$C_0^j = \epsilon^S \frac{A}{t} \quad \text{Single layer capacity}$$

$$C_0 = \sum_{j=1}^L C_0^j = L \epsilon^S \frac{A}{t} \quad \text{Total capacity}$$



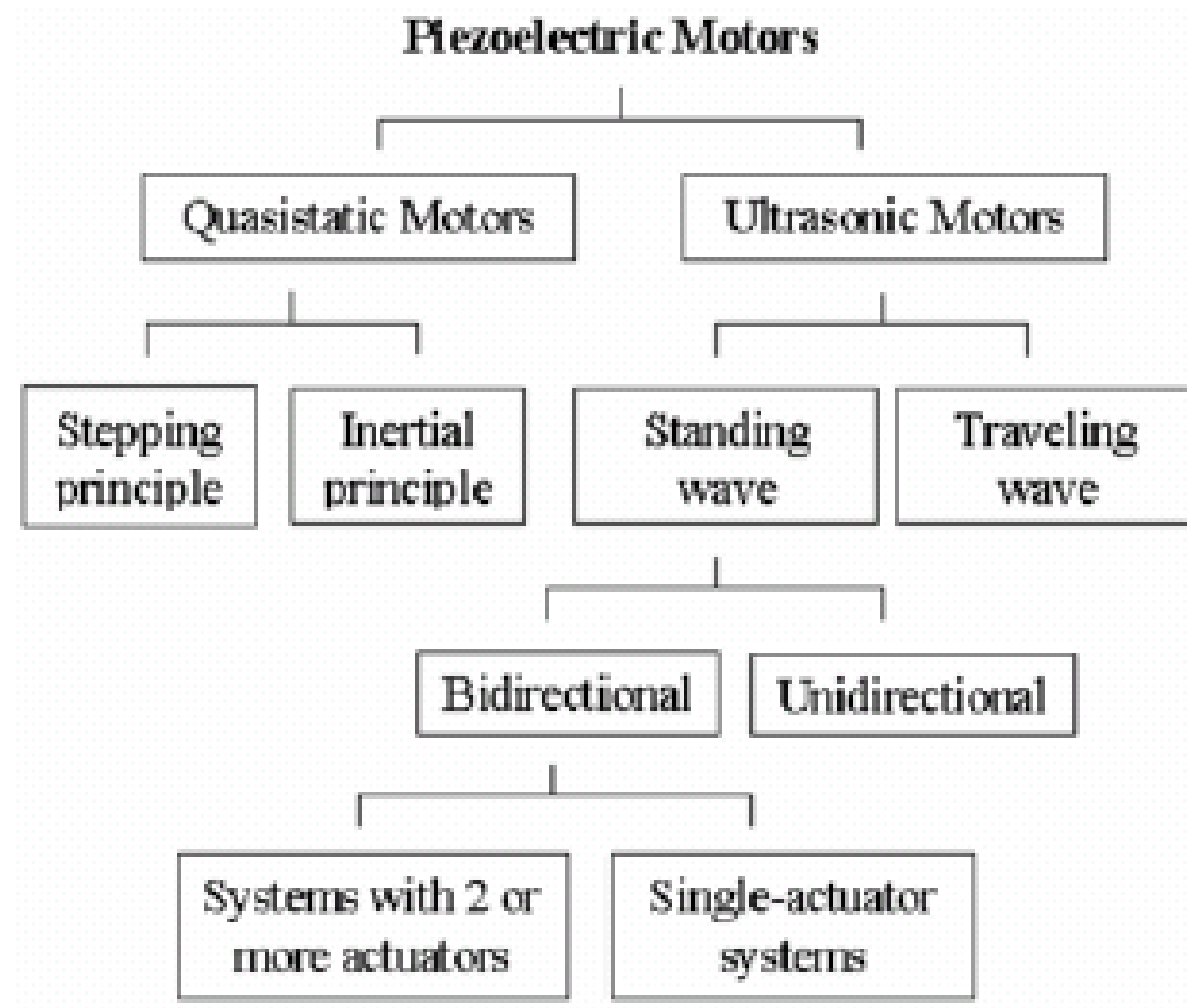
Stack

Example:

<https://www.piceramic.com/en/products/piezoceramic-actuators/linear-actuators/>



Piezo-motors classification



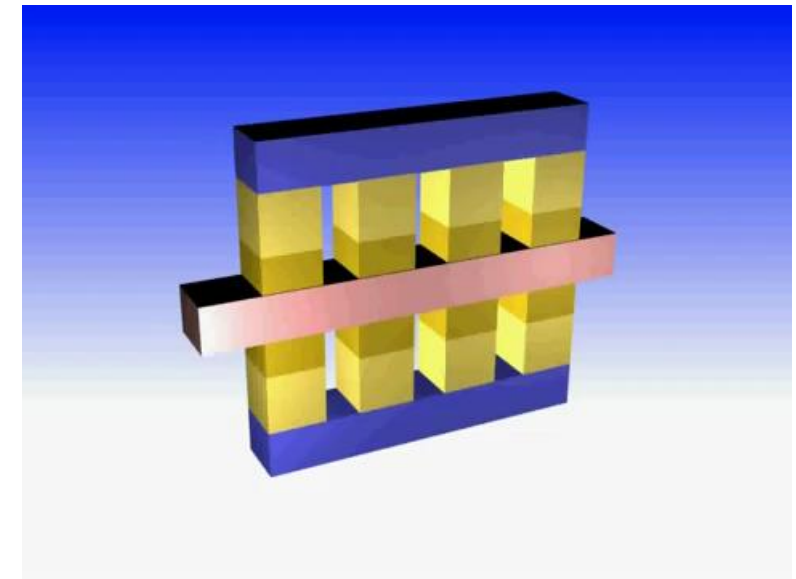
Combination of standing waves



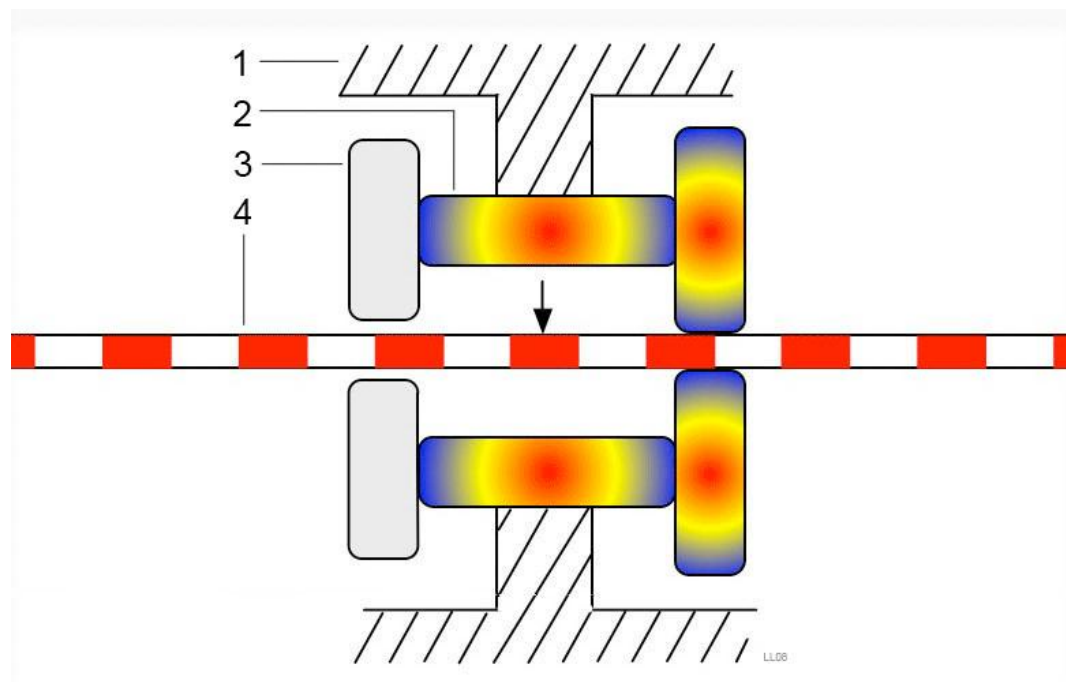
Non-resonant (quasi-static) piezoelectric actuators

Stepping

The rotor or slider is passed hand-over-hand between two sets of clamping and driving actuators. At any point in time, the set of actuators that is moving in the drive direction is the set that is clamped to the slider or rotor, while the other set is unclamped and moving backwards, preparatory to taking a new bite.



Inchworm



Three independently driven piezoelectric ceramics

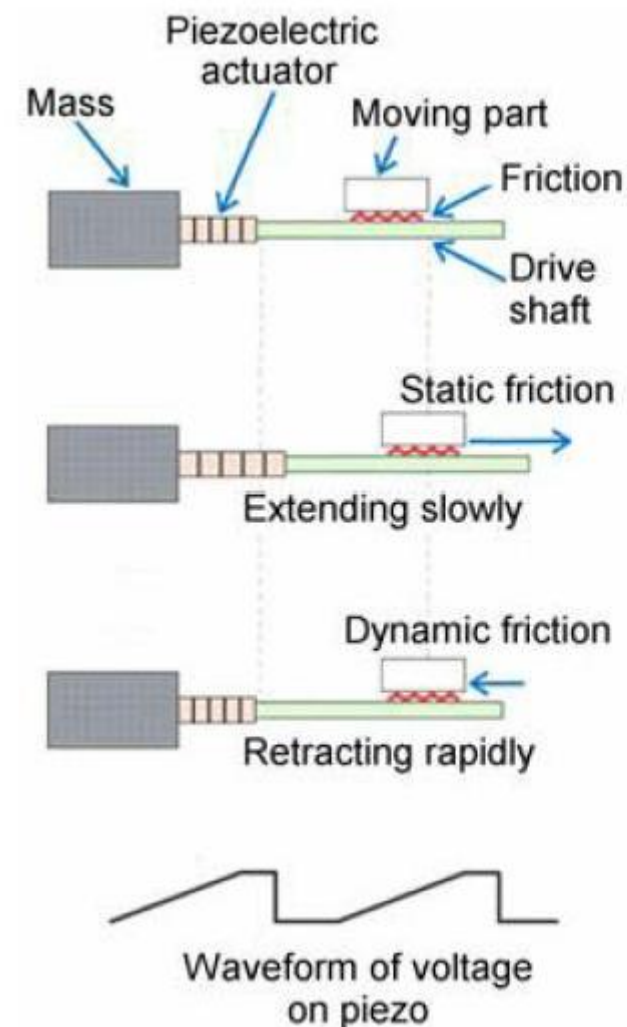
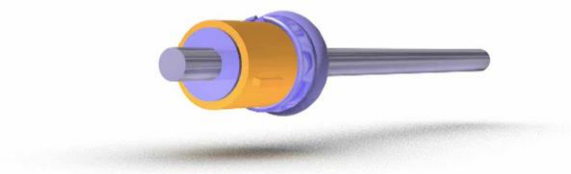
1. The first ceramic (1) clamps the rotor.
2. (2) is actuated in axial mode.
3. (3) is actuated as (1).
4. (1) is relaxed releasing the rotor.
5. (2) is relaxed displacing axially the rotor.
6. (3) is relaxed.



Non-resonant (quasi-static) piezoelectric actuators

Inertial

Inertia and the difference between the static and dynamic friction coefficients is used (the frictional force on the slider is reduced when slippage generated by the sliders' inertia occurs). During the slower displacement of the actuator in the drive direction, the slider follows the actuator with no slippage. In the next phase, the actuator moves in the opposite direction so rapidly that the slider, unable to follow due to its inertia, lets the actuator slip back beneath it.



Actuator is composed of a piezoelectric actuator, a drive shaft and a moving part.

To drive moving part to the right, the voltage applied to the piezo is increased slowly. The moving part follows under static friction.

Then the voltage is decreased rapidly; the moving part slips, experiencing dynamic friction for a shorter time.

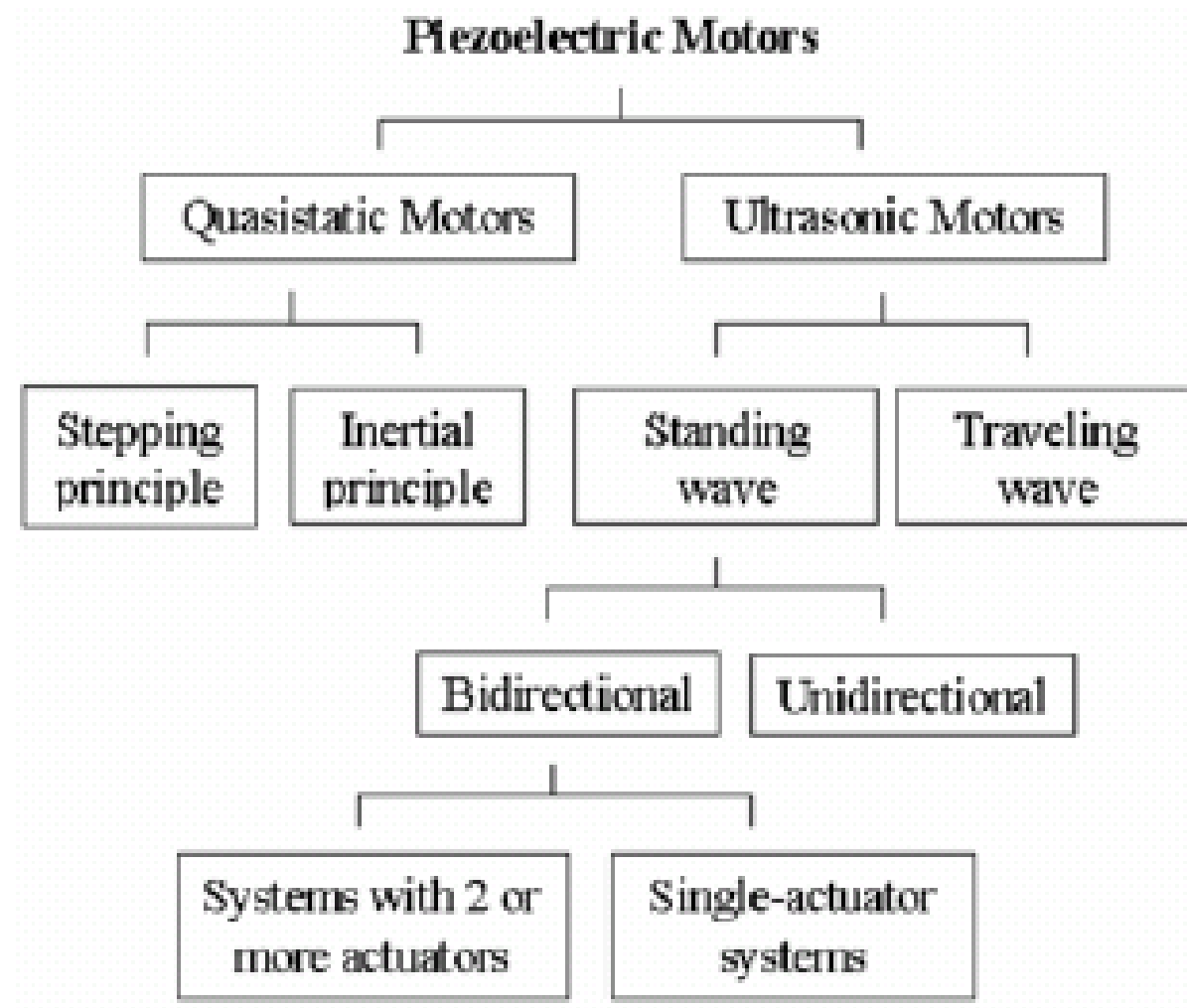
To reverse motion, the waveform is mirrored, rising rapidly and falling slowly.



PI



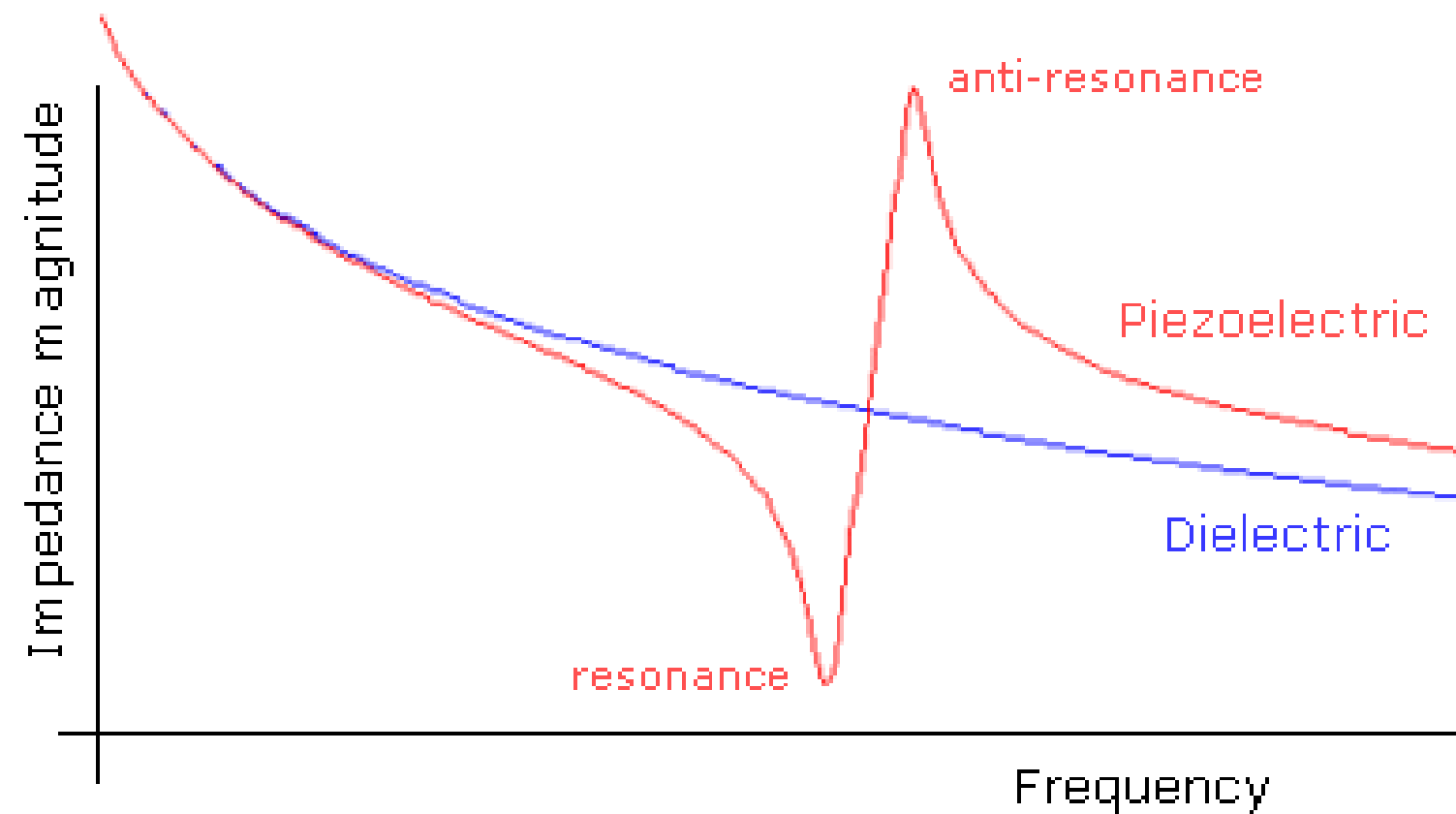
Piezo-motors classification



Combination of standing waves

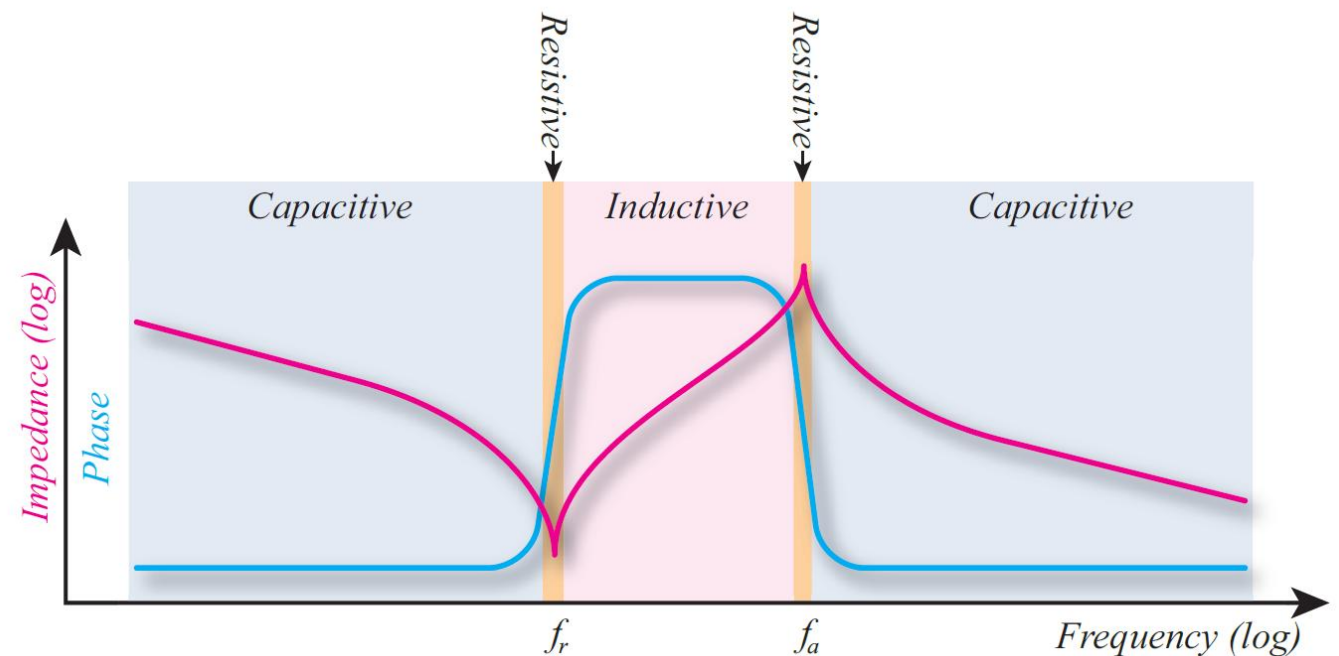
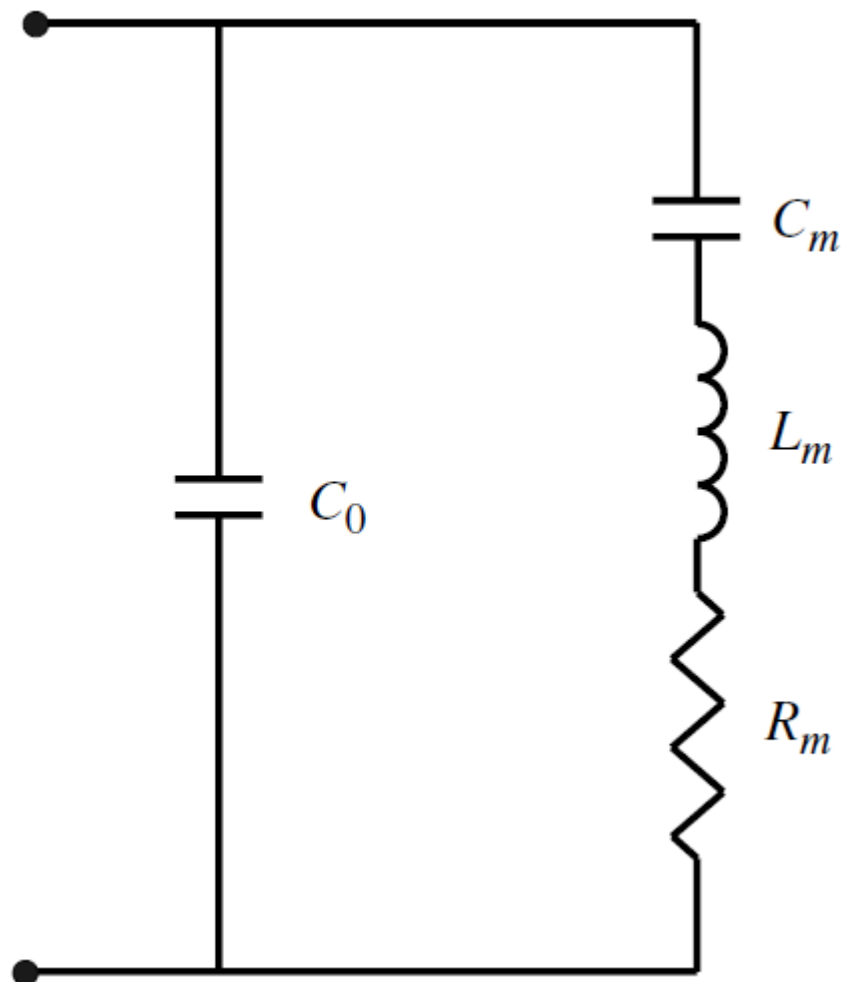


Resonant (ultrasonic) piezoelectric actuators



Resonant (ultrasonic) piezoelectric actuators

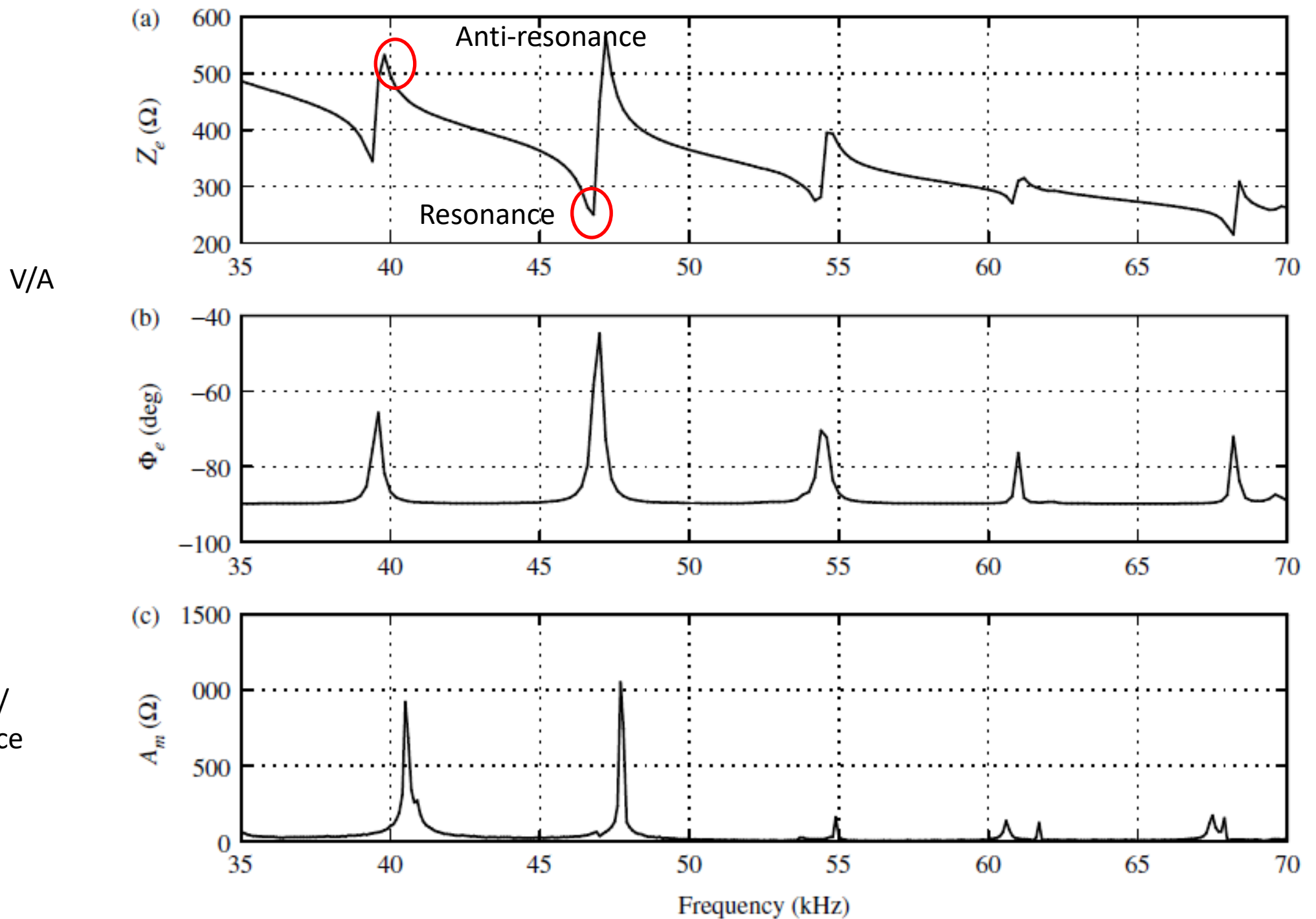
Equivalent circuit



- C_0 Clamped capacitance – capacitance of the piezoelectric ceramic - Stator
 - C_m Motional impedance
 - L_m Motional inductance
 - R_m Motional resistance
- } Rotor properties



Resonant (ultrasonic) piezoelectric actuators



Resonant (ultrasonic) piezoelectric actuators

Operation point

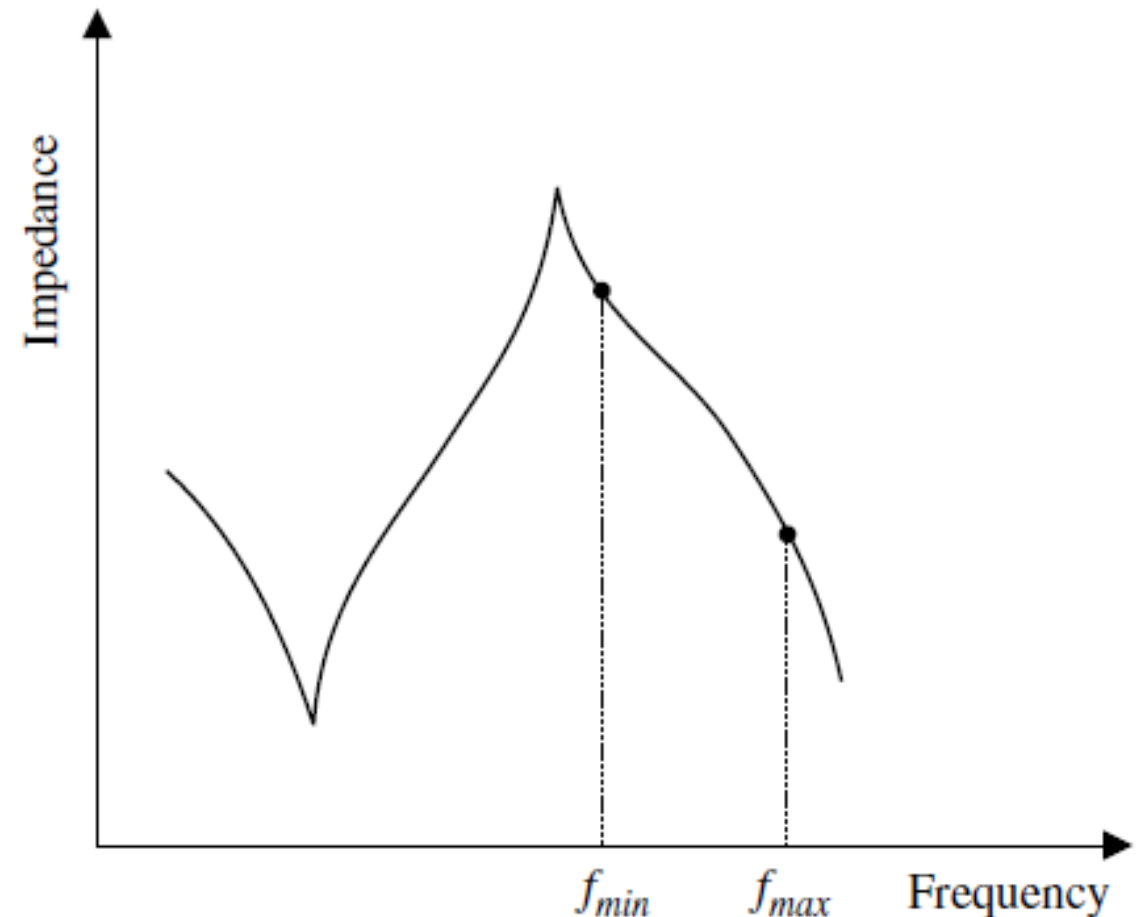
Quality factor

$$Q = \frac{\sqrt{L_m / C_m}}{R_m}$$

Resonant piezoelectric actuators are usually driven in a frequency range close to the antiresonance frequency of the device.

Antiresonance modes (B-type modes) are better for actuators since their quality factor is higher than resonance modes over the entire range of vibration frequency.

The smoothly decreasing impedance curve after antiresonance is therefore used to define the range of frequencies allowed for motor speed control.



Resonant (ultrasonic) piezoelectric actuators

How to pass from resonance to macroscopic motion?

Frictional mechanism between the stator and the rotor:

- 1- the vibrational motion is **transformed** to a microscopic elliptic motion of the stator;
- 2- this elliptic motion is **transmitted** to the rotor through friction or impact mechanisms

$$y = A_y \sin(\omega_y t + \varphi_y)$$

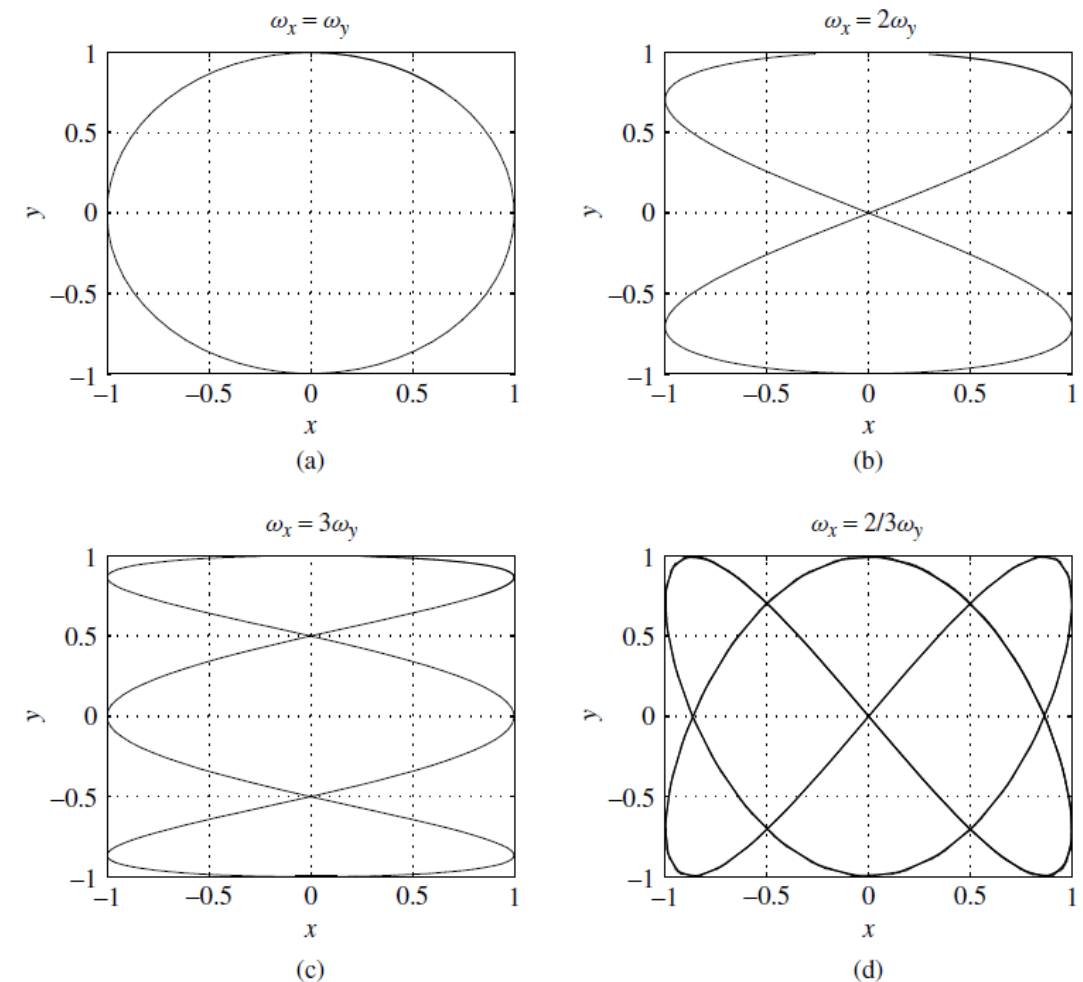
$$x = A_x \sin(\omega_x t + \varphi_x)$$

The first step is usually based on a **combination of two linear motions on orthogonal axes**. The combination of harmonic motion in two mutually orthogonal directions leads to the well-known *Lissajous loci*

PHASE = 90°

The practical application to piezoelectric resonant drives involves either **the combination of two orthogonal vibration modes** at the same frequency or the use of what is known as **the mode conversion principle**

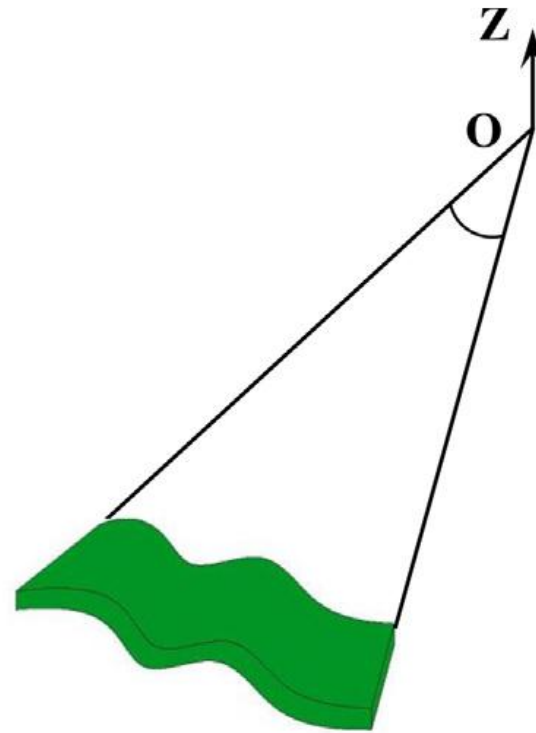
LINEAR OR ROTATIONAL MOTORS



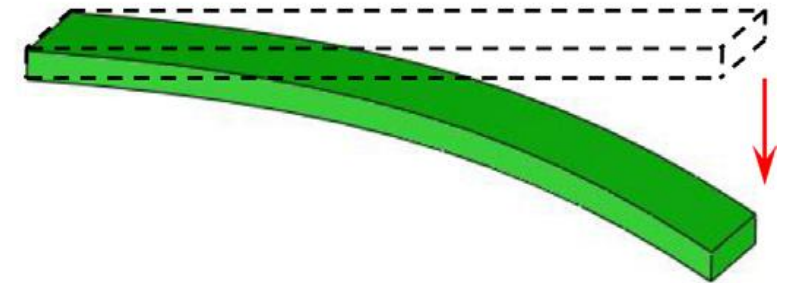
Resonant (ultrasonic) piezoelectric actuators

Standing waves

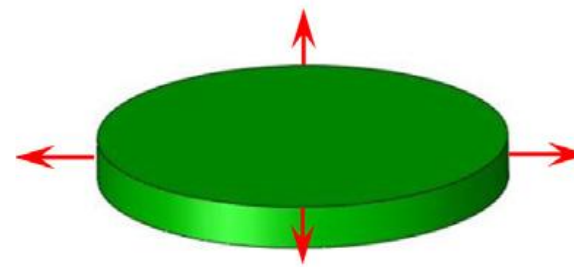
- Flexural Vibration Mode (a)
Pure bending in circumferential direction
- Bending Vibration Mode (b)
- Radial Vibration Mode (c)
- Longitudinal Vibration Mode (d)
- Torsional Vibration Mode (e)



(a)



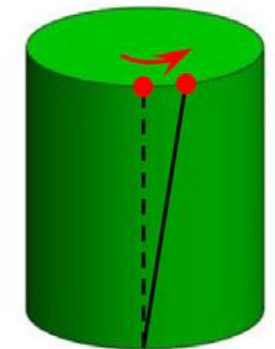
(b)



(c)



(d)



(e)



Resonant (ultrasonic) piezoelectric actuators

Combination of different modes to achieve elliptic motion

- Motors using resonance modes of the same type (at the same resonance frequency)
- Motors using resonance modes of different types at the same frequency
- Motors using resonance modes of different types at different frequencies
- Motors using mode separation (same modes, same frequencies – nonsymmetric geometry)
- Motors using mode conversion (single longitudinal vibration mode + impact)



Resonant (ultrasonic) piezoelectric actuators

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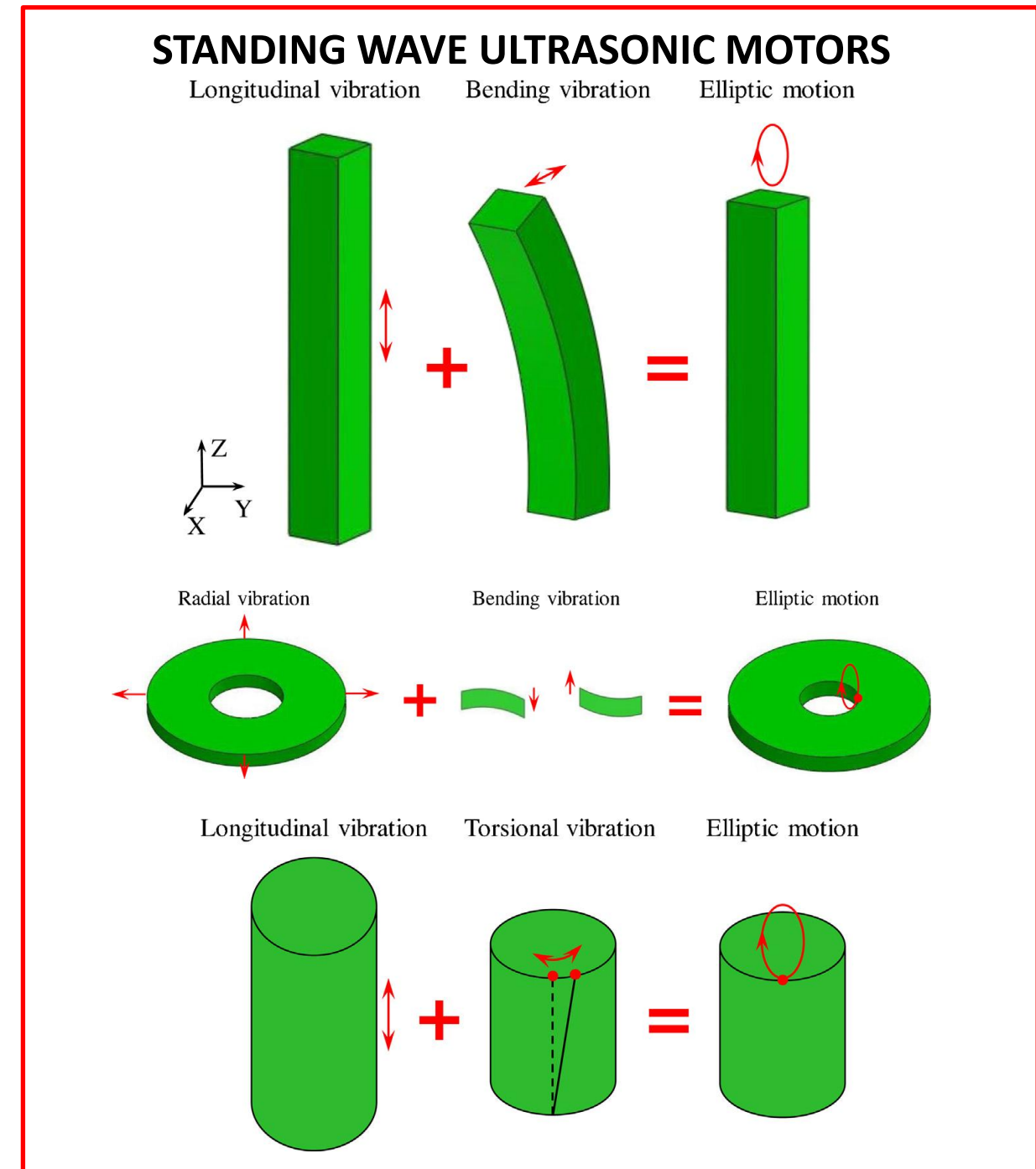
**TRAVELLING WAVE
ULTRASONIC MOTORS**



Resonant (ultrasonic) piezoelectric actuators

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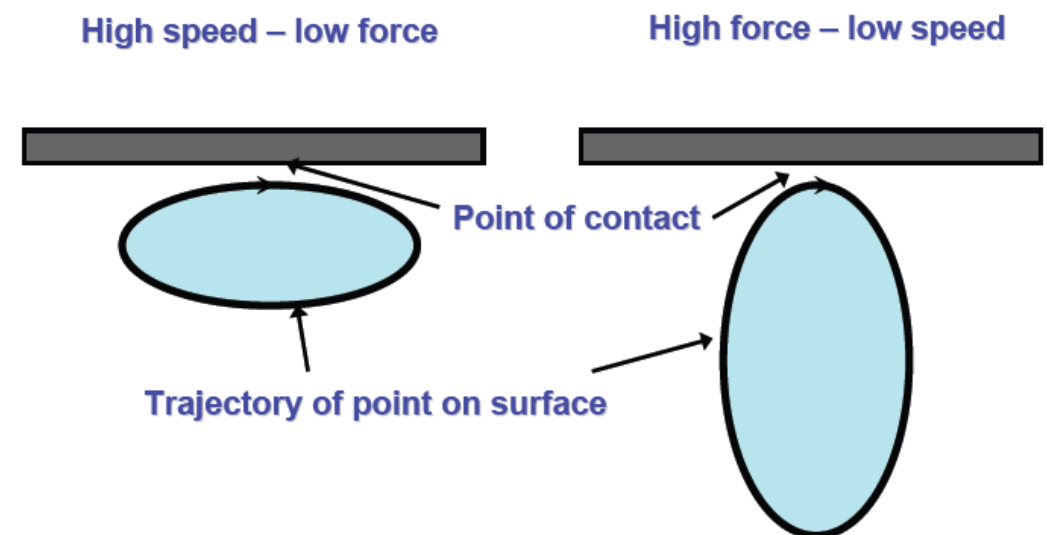
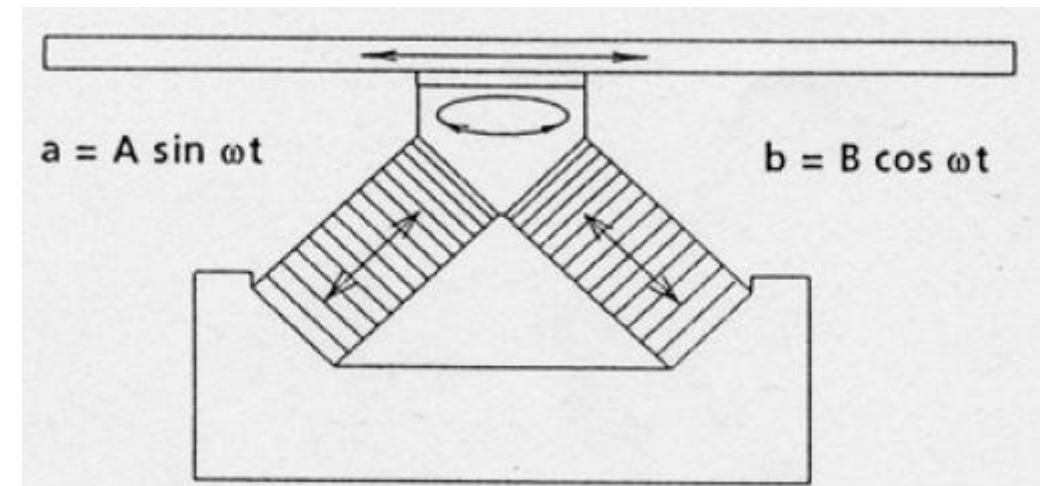


Resonant (ultrasonic) piezoelectric actuators

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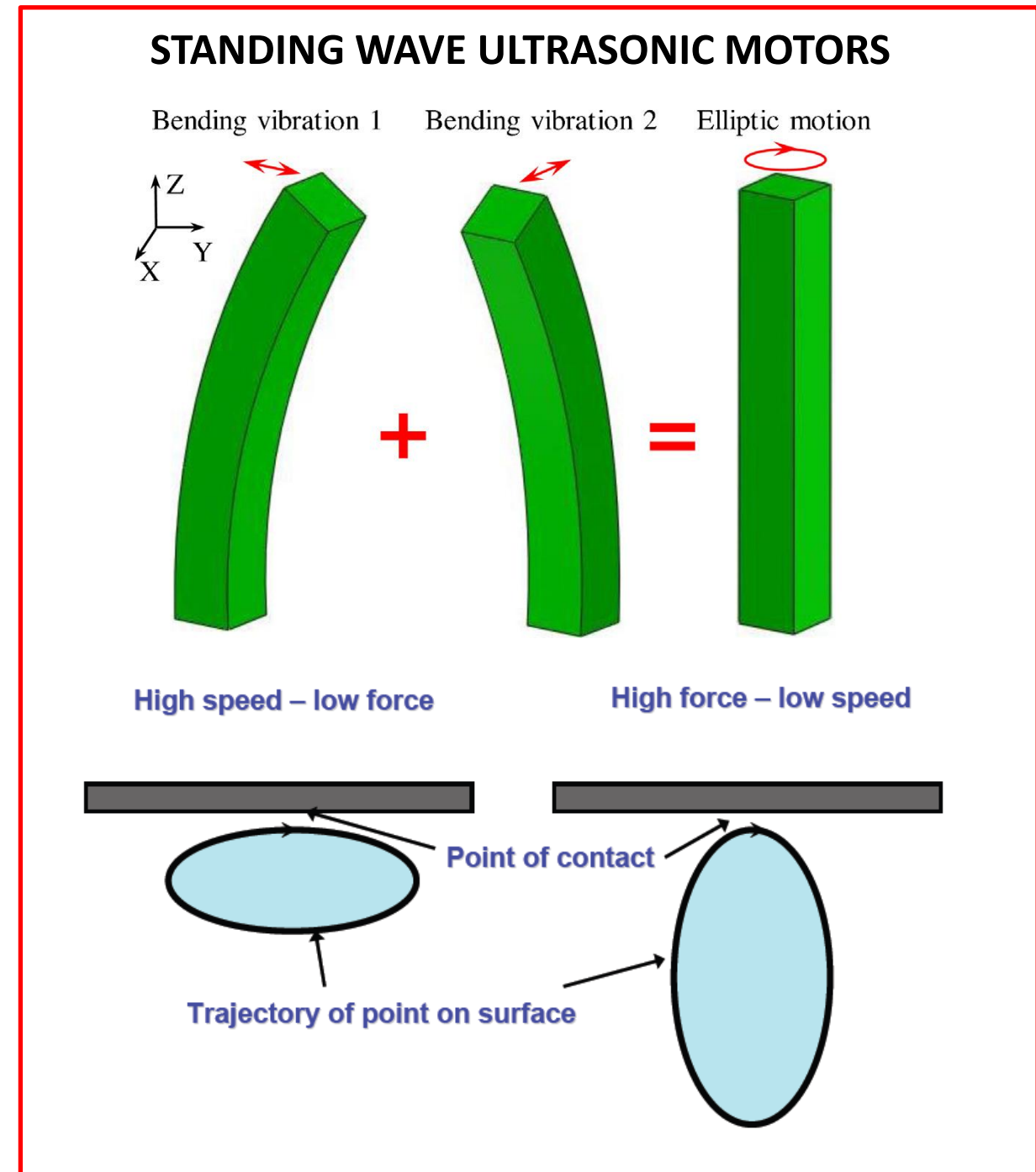
STANDING WAVE ULTRASONIC MOTORS



Resonant (ultrasonic) piezoelectric actuators

Combination of different modes to achieve elliptic motion

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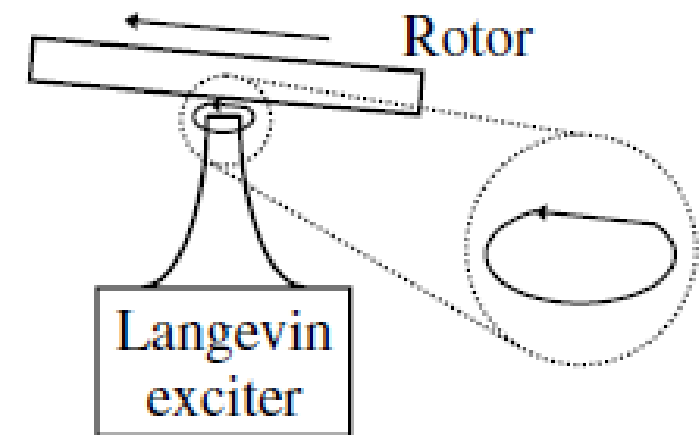


Resonant (ultrasonic) piezoelectric actuators

Combination of different modes to achieve elliptic motion

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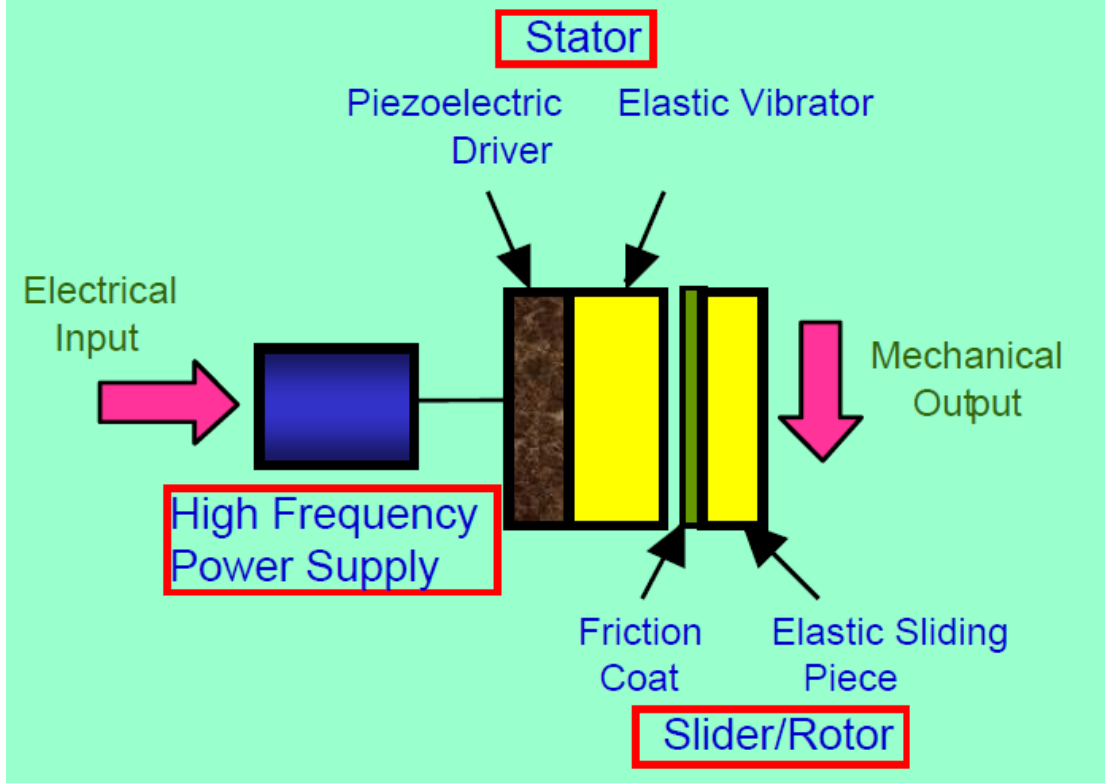
STANDING WAVE ULTRASONIC MOTORS



Resonant (ultrasonic) piezoelectric actuators

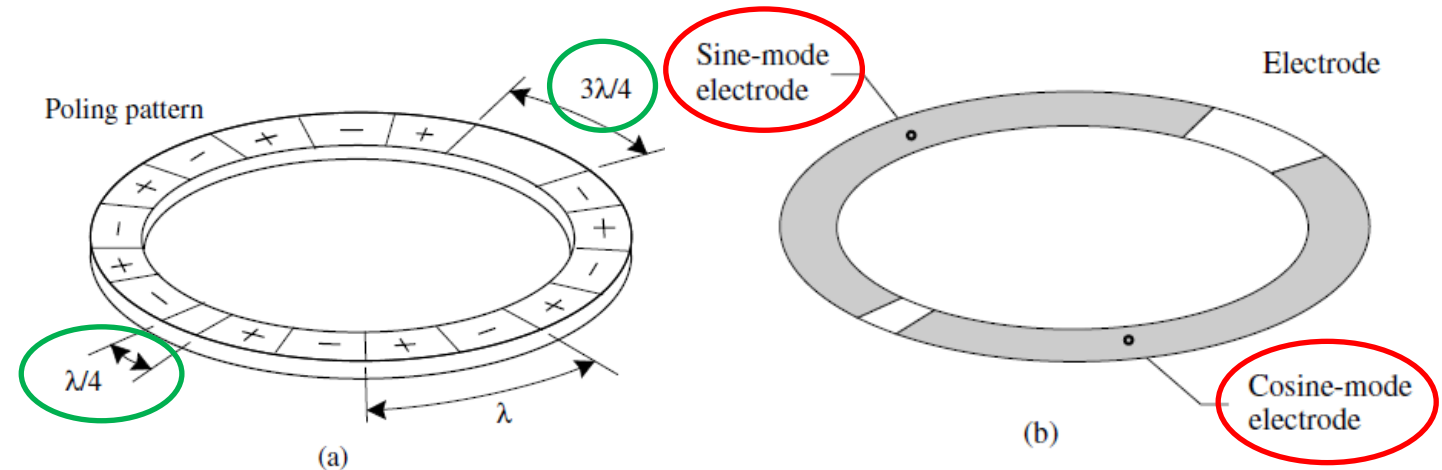
Travelling Wave Ultrasonic Motor (TWUM)

Fundamental construction of an ultrasonic motor



Ultrasonic (20kHz)

Sum of two standing waves with
90° phase difference (in **time** and **space**)

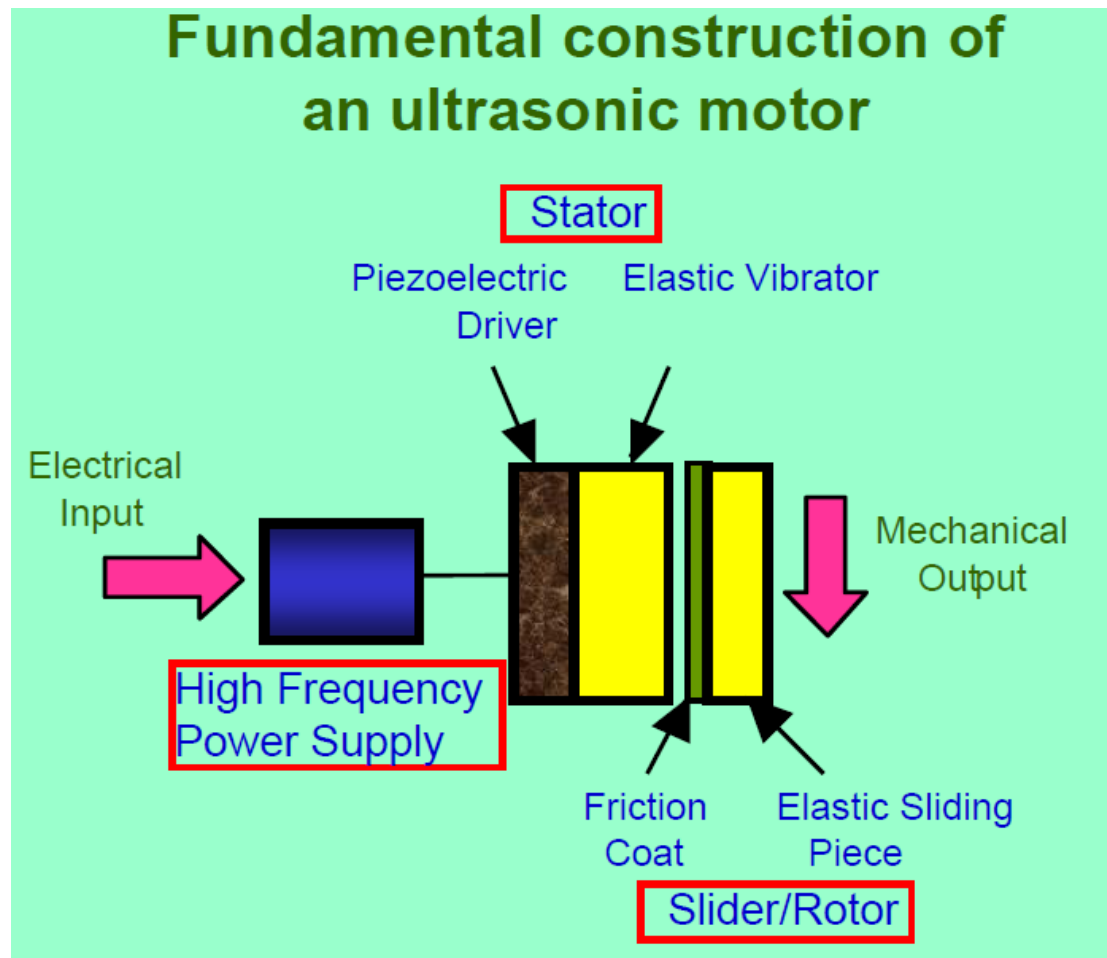


$$u(\phi, r, t) = AR(r) \cos(k\phi) \cos(\omega t) \mp AR(r) \sin(k\phi) \sin(\omega t)$$



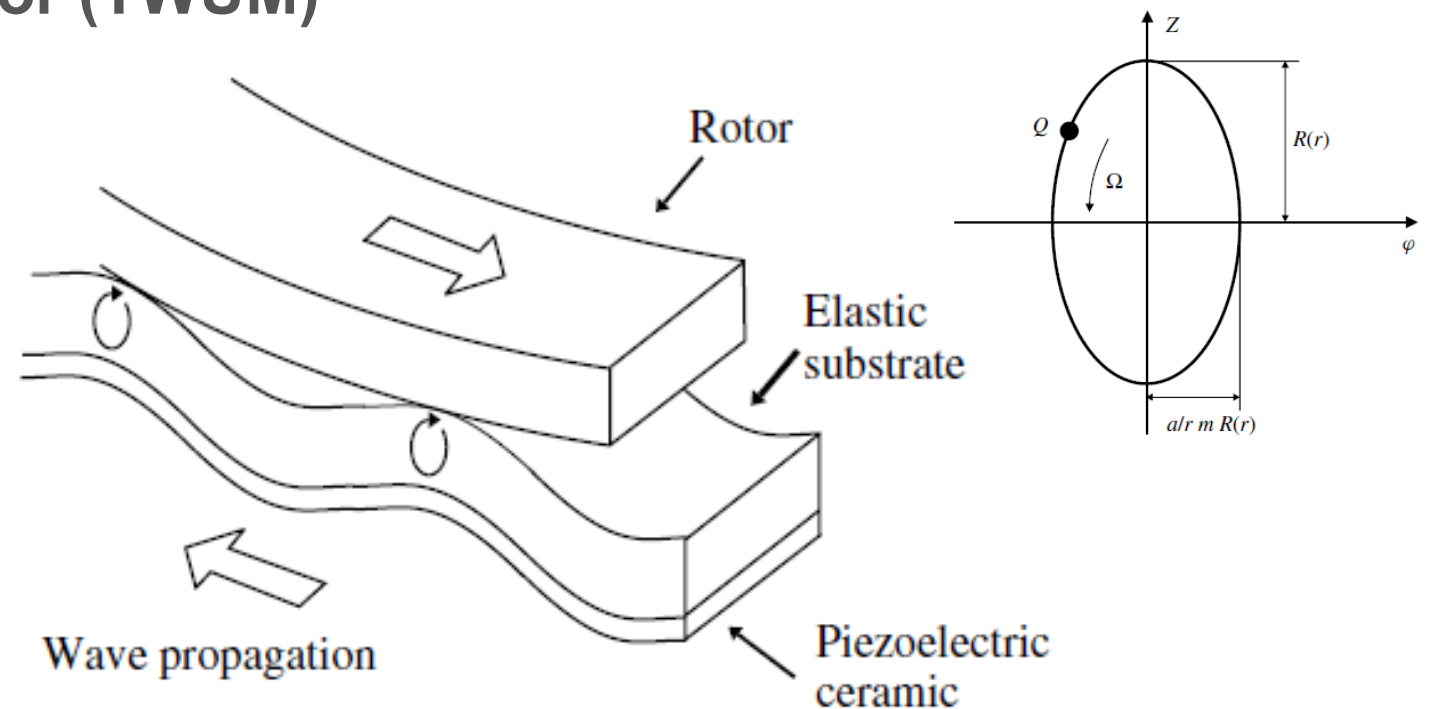
Resonant (ultrasonic) piezoelectric actuators

Travelling Wave Ultrasonic Motor (TWUM)



Ultrasonic (20kHz)

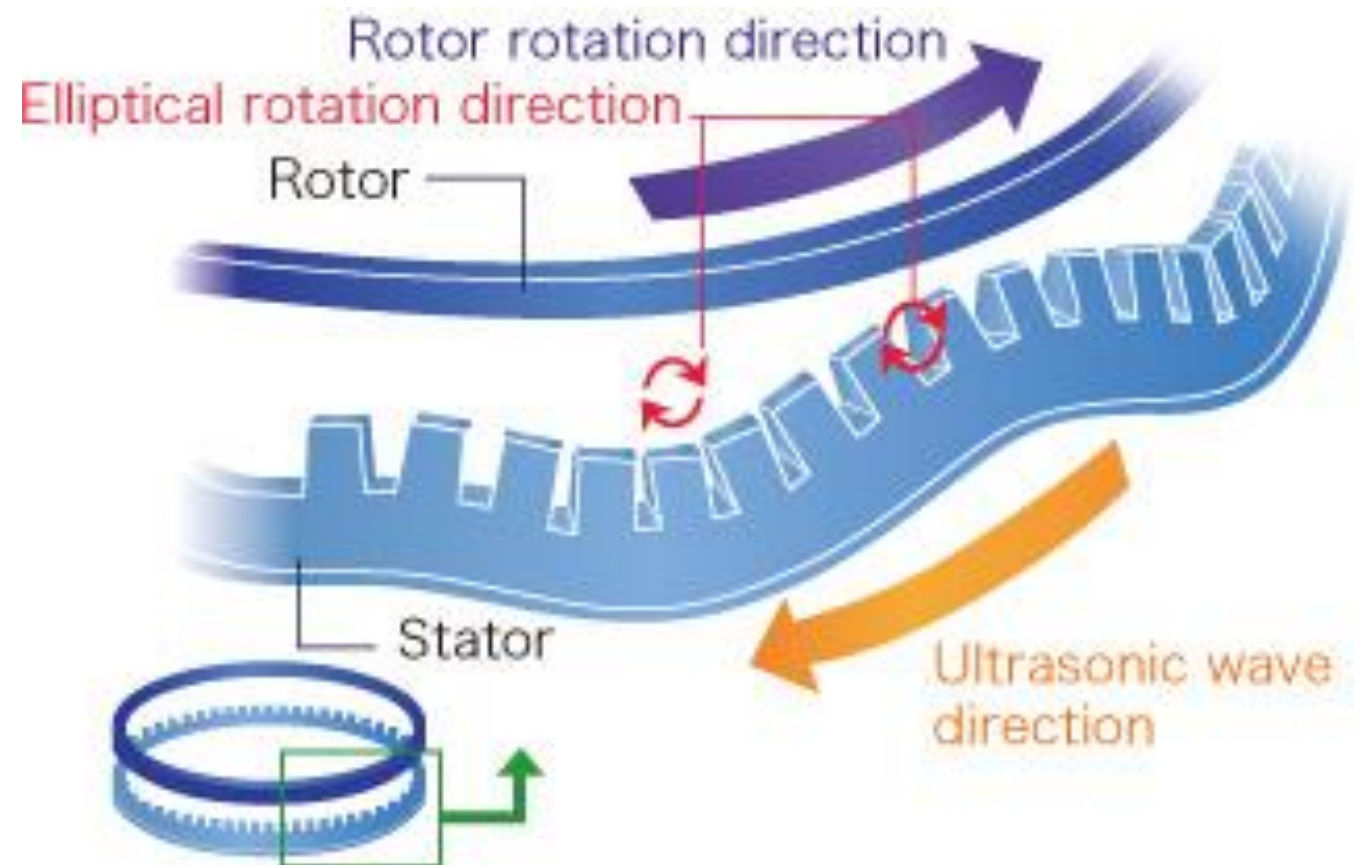
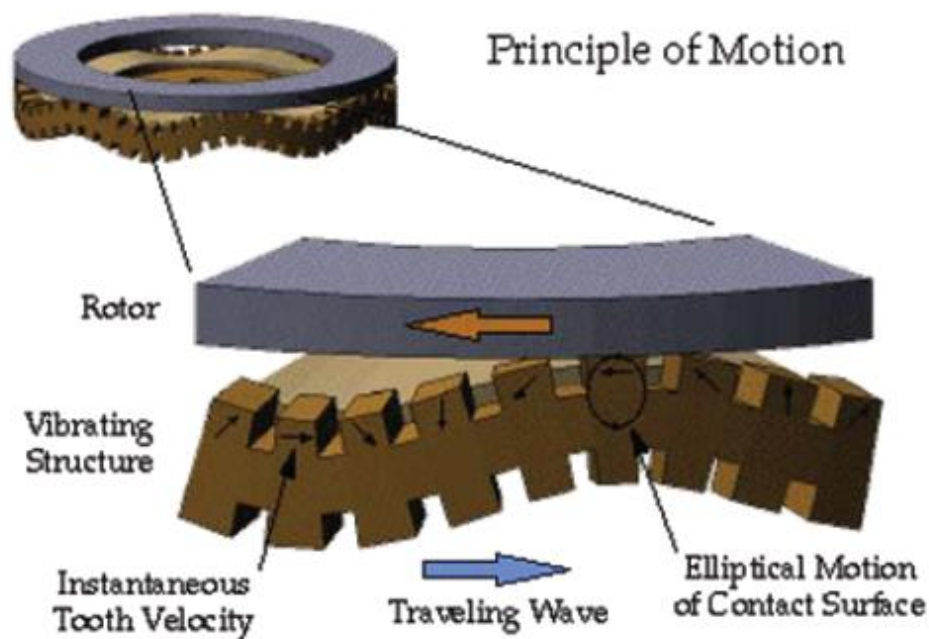
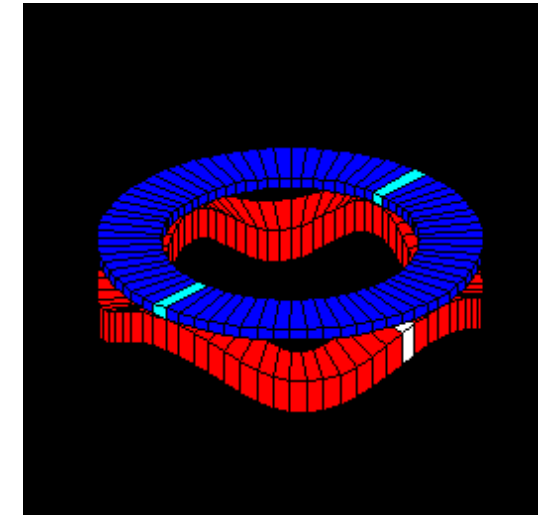
Sum of two standing waves with 90° phase difference (in **time** and **space**)



Resonant (ultrasonic) piezoelectric actuators

ROTATIONAL - Travelling Wave Ultrasonic Motor (TWUM)

- A traveling wave is induced on the stator.
- The direction can be reversed by reversing the excitation.
- The stator includes teeth that magnify the displacement.



Resonant (ultrasonic) piezoelectric actuators

ROTATIONAL - Travelling Wave Ultrasonic Motor (TWUM)



Figures of merit and scaling trends

	Nonresonant actuators			Resonant actuators
Figures of merit	Stack	Benders	Inchworm	TWUM
Force, F	$10^2\text{--}10^5$ N	$10^{-2}\text{--}10^2$ N		Torque $\leq 10^4$ Nm
Displacement, S	0.1–0.6% (dynamic up to 1.7%)	$10^{-2}\text{--}10$ mm	Unlimited	Unlimited
Work density, W_V	$\approx 10^{-2}$ J/cm ³	$10^{-4}\text{--}10^{-2}$ J/cm ³	$\approx 10^{-3}$ J/cm ³	$10^{-3}\text{--}10^{-2}$ J/cm ³
Power density, P_V	$10^2\text{--}10^4$ W/cm ³	$10^{-1}\text{--}10^1$ W/cm ³	$\approx 10^{-4}$ W/cm ³	$\approx 10^{-1}$ W/cm ³
Bandwidth, f	$\leq 10^5$ Hz	$\leq 10^3$ Hz	$\leq 10^2$ Hz	$\leq 10^2$ Hz
Efficiency, η	70–90%			
Scaling trends				
<i>Force</i>	$F \propto L^2$			
<i>Stroke</i>	$S \propto L$			
<i>Work per cycle</i>	$W \propto L^3$			
<i>Energy density</i>	$W_V \propto L^0$			
<i>Bandwidth</i>	$f \propto L^{-1}$			
<i>Power density</i>	$P_V \propto L^{-1}$			



Application

Tremor Compensation

Consumer Electronics

Digital Camcorder
(Panasonic et al.)

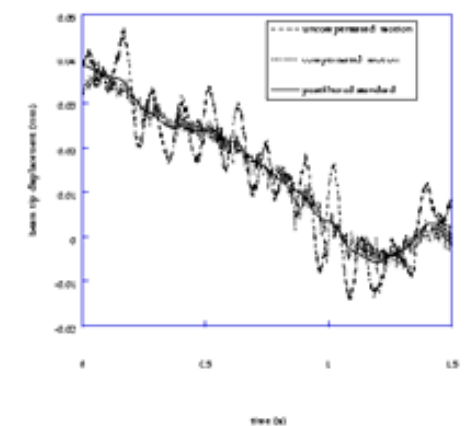
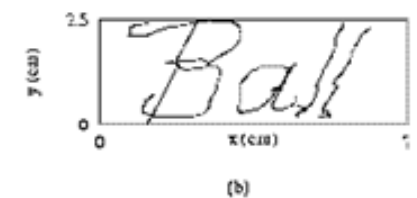
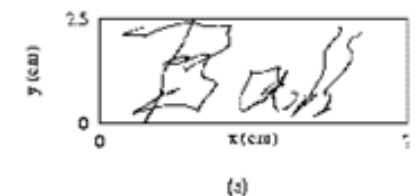


Compensation Strategy:

The instrument detects its own motion, distinguishes between desired and undesired motion using **active filtering and inertial sensors**, and actively compensates for undesired motion by an equal but opposite deflection of its own tip.

Microsurgical device

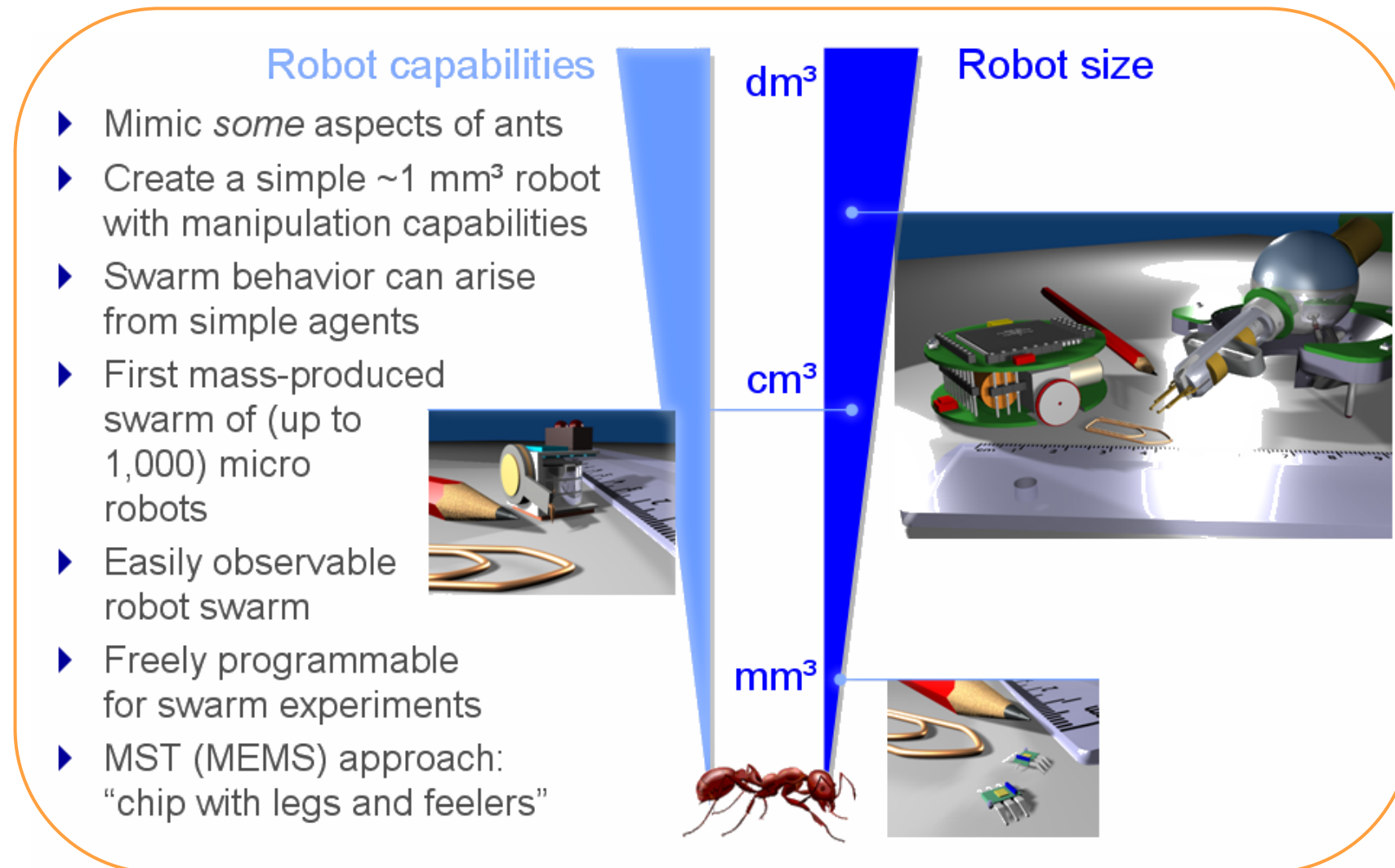
Intelligent hand-held instrument for ophthalmologic microsurgery
(MRCAS - Carnegie Mellon Univ. Robotics Institute - Pittsburgh, PA)



Application

I-SWARM Project

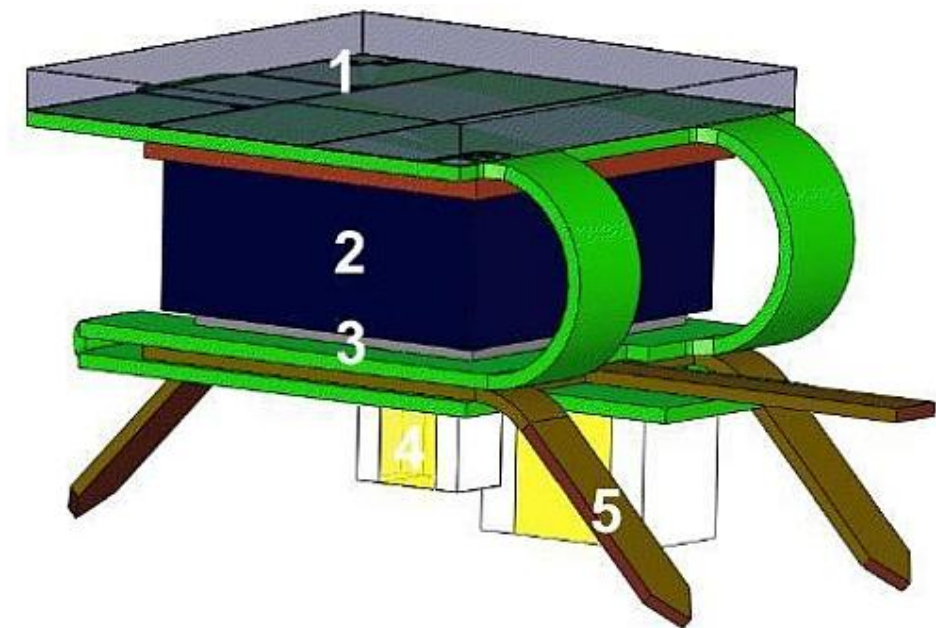
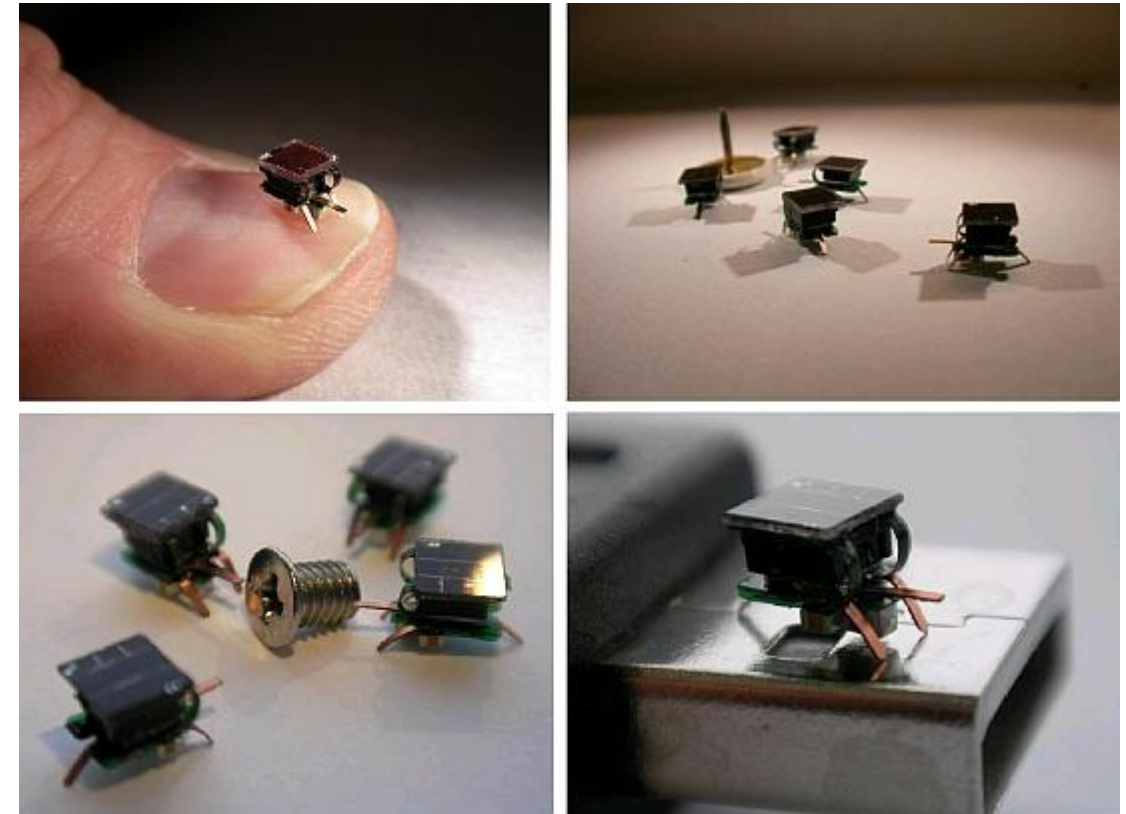
Cluster and swarm of minirobot for precision manipulation and/or cooperation



Application

I-SWARM Project

**LOCOMOTION UNIT BASED ON
PVDF PIEZOELECTRIC FILM
“MICROLEGS”**

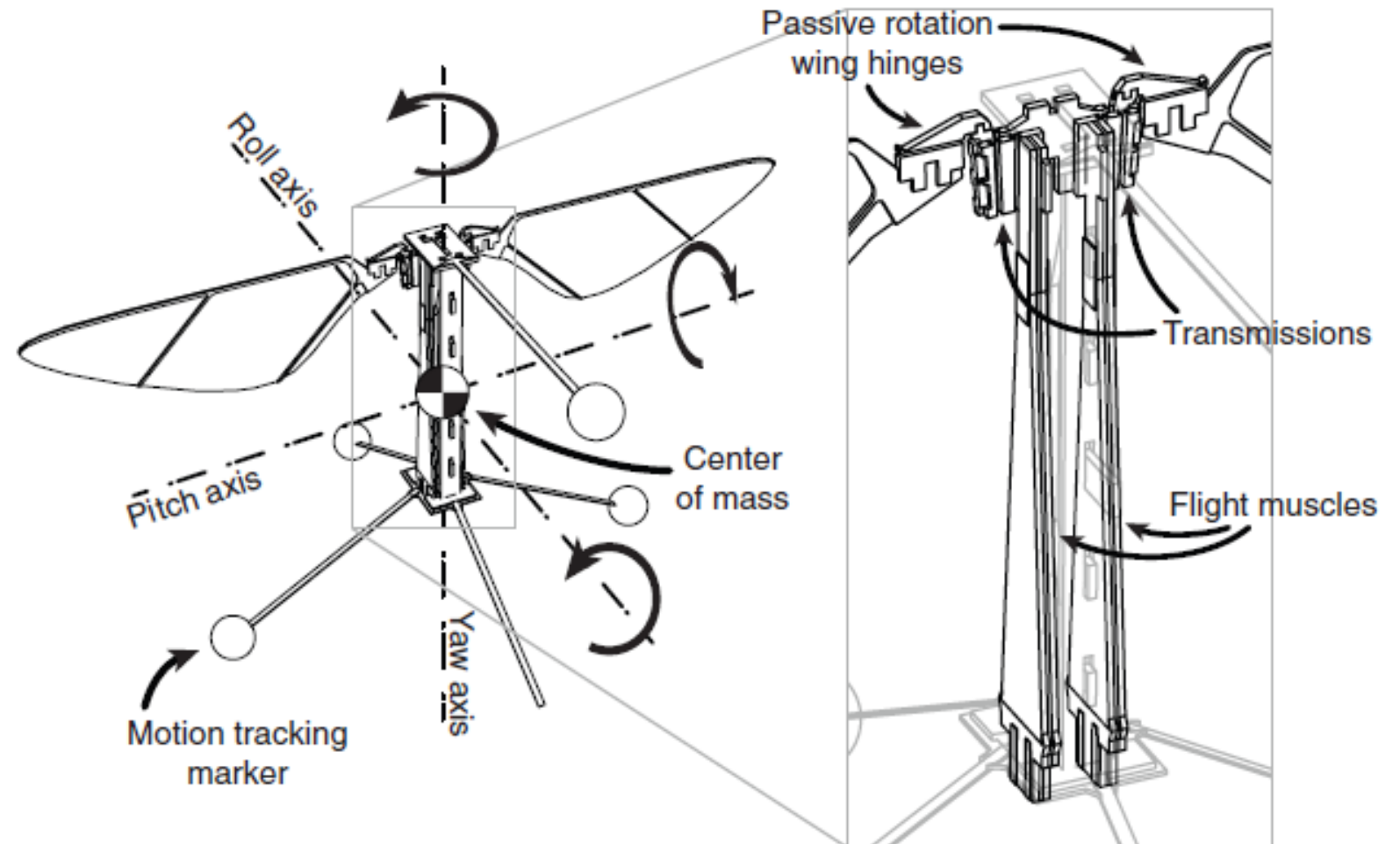
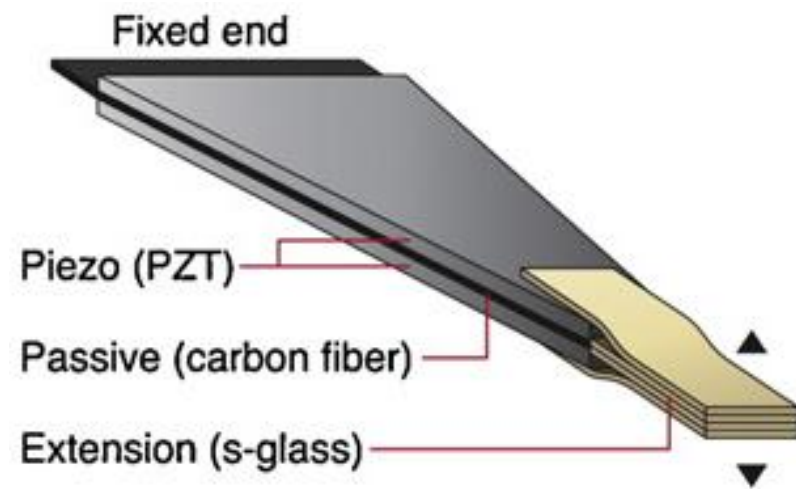
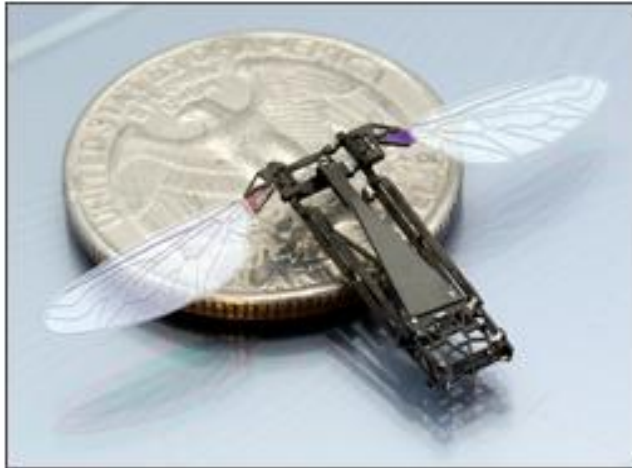


(1) solar cell, (2) IR-communication module, (3) an ASIC,
(4) capacitors, (5) locomotion module



Application

Flying insect



M. Karpelson, et al., Driving high voltage piezoelectric actuators in microrobotic applications, Sens. Actuators A: Phys. (2012)



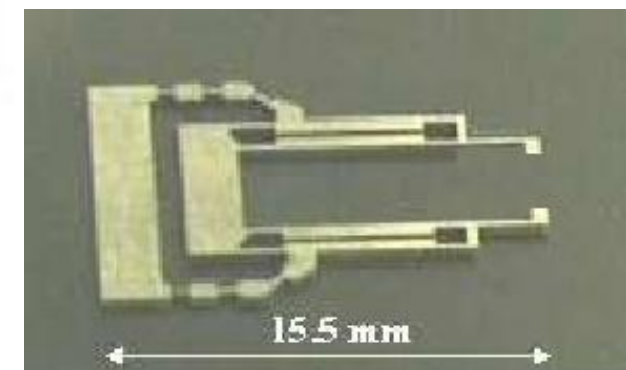
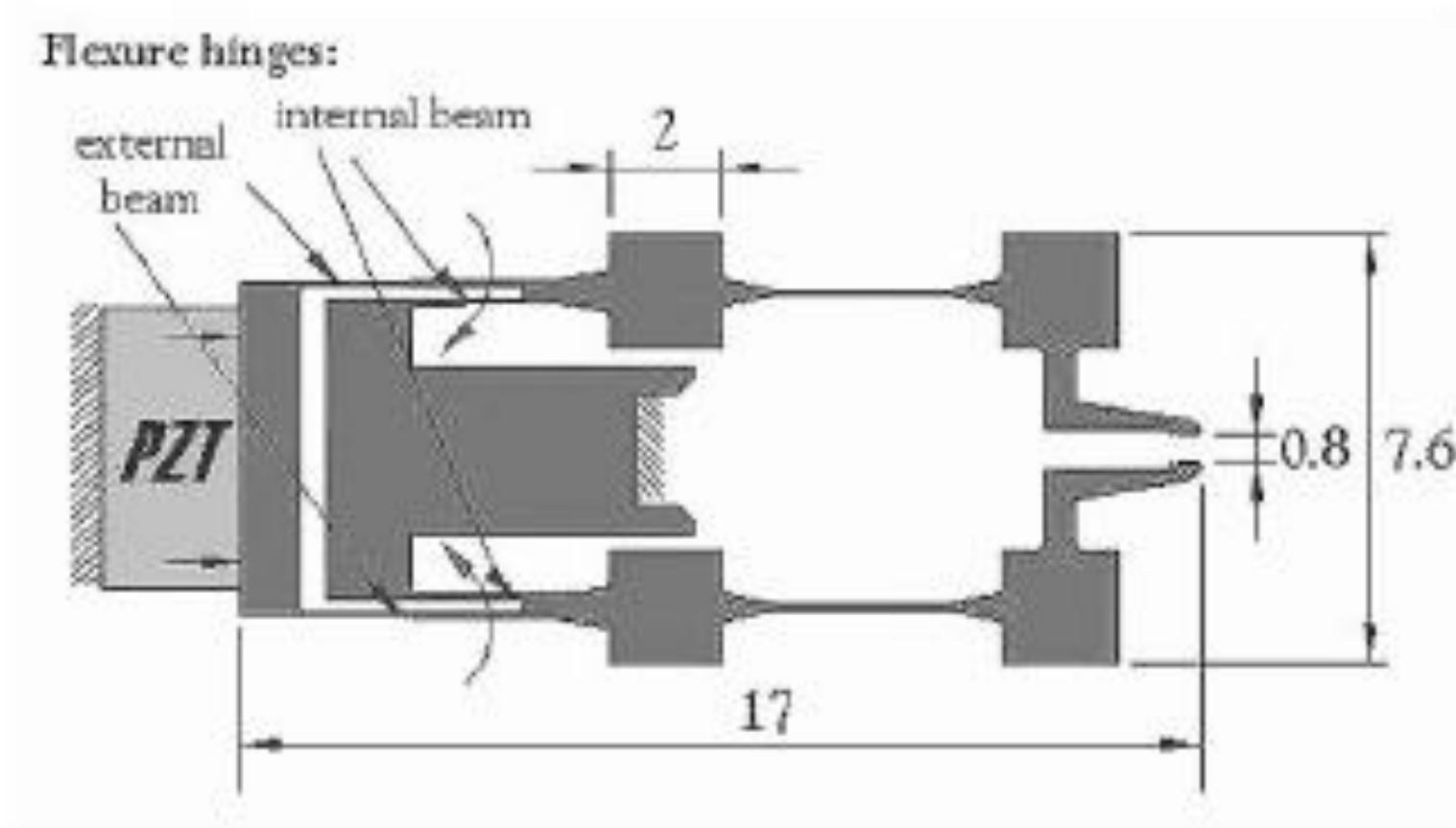
Application

Flying insect



Application

Displacement amplification mechanism



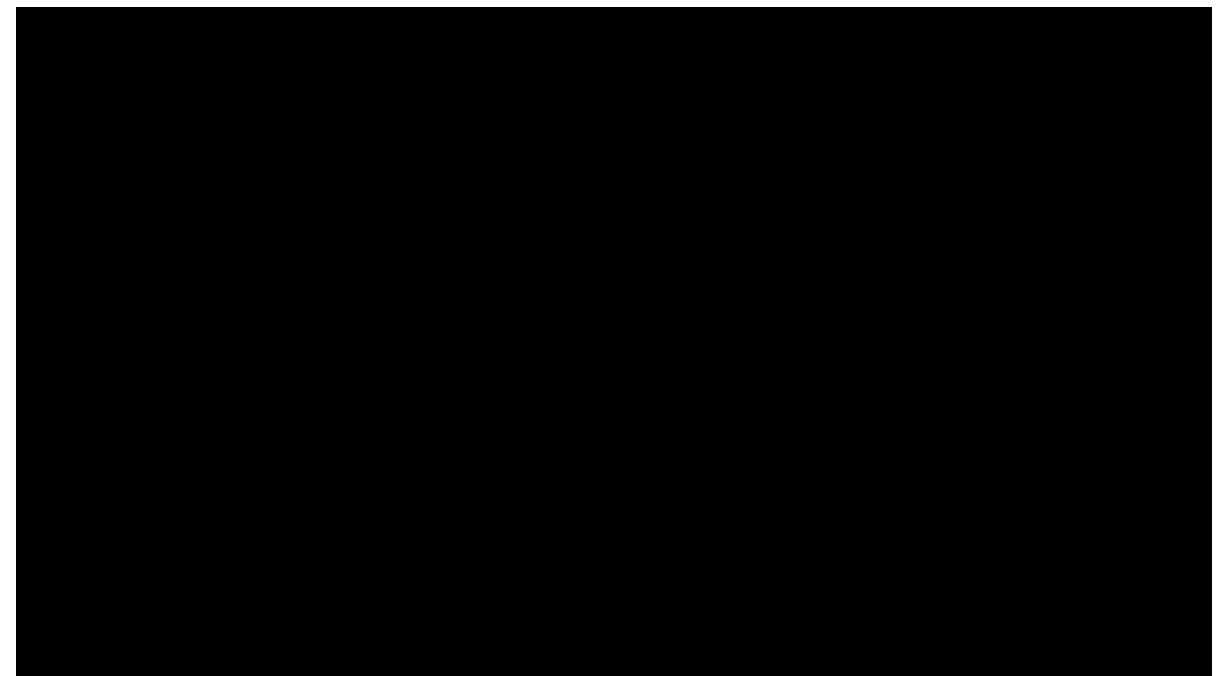
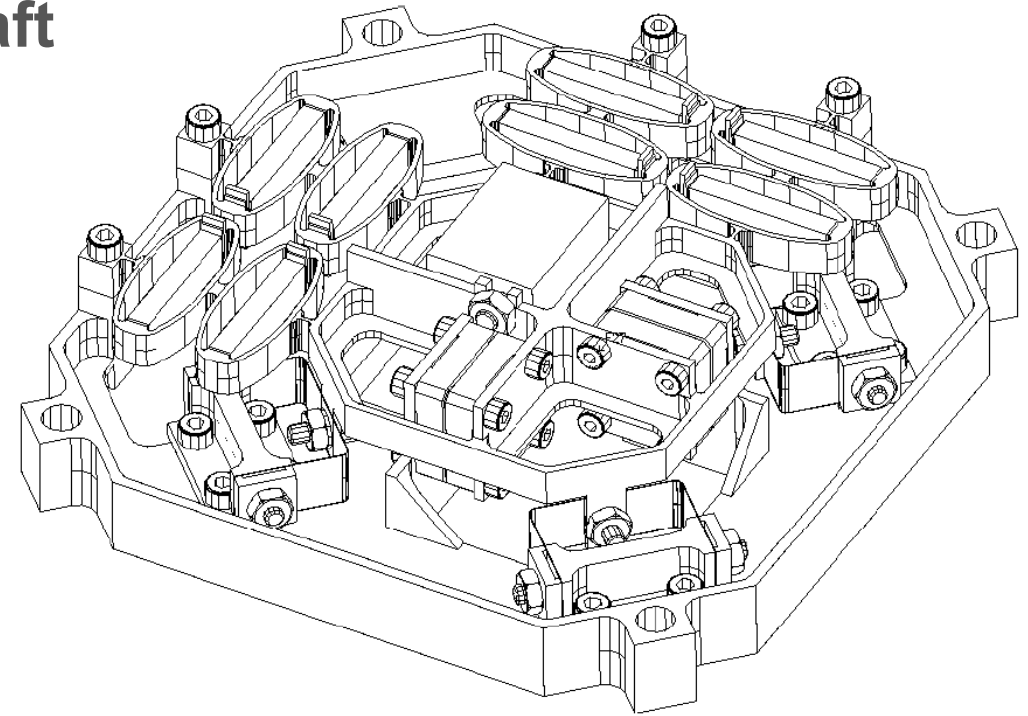
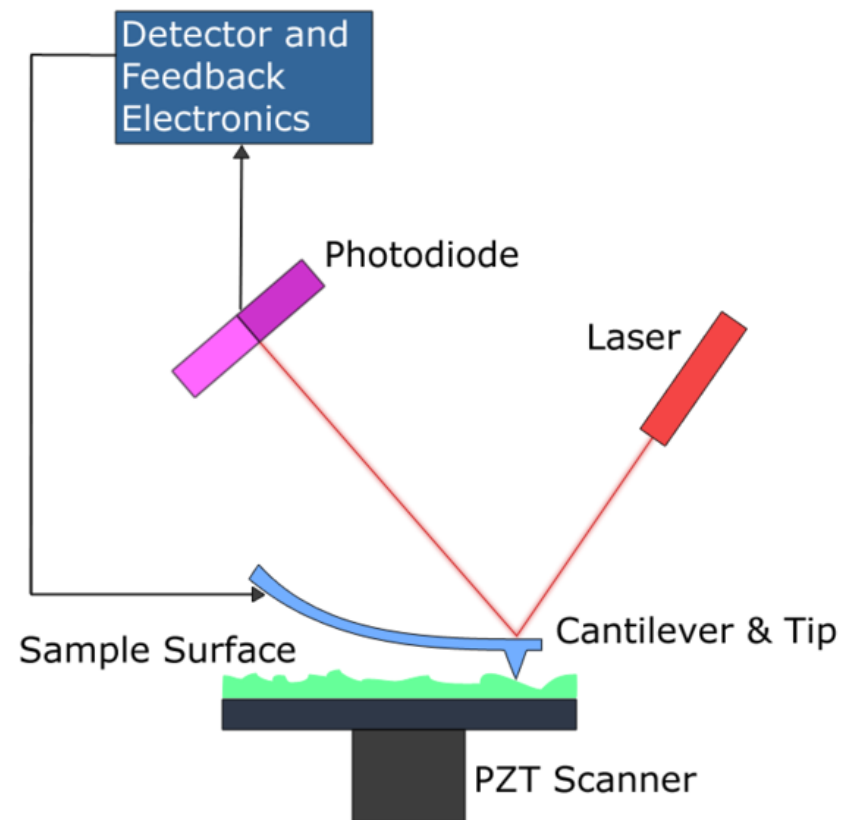
Mini-plier for assembly (laser machining)



Application

Scanning mechanism for AFM on a spacecraft

Atomic Force Microscope



Application



Nanomotor® Version	S=Small	L=Long	B=Big
Length of Nanomotor (5 mm vanish in mounting)	18.1 mm	31.4 mm	31.6 mm
Length of standard mounting	13 mm	13 mm	13 mm
Length of Nanomotor incl. standard mounting	26.1 mm	39.4 mm	39.6 mm
Diameter of Nanomotor	3.3 mm	3.3 mm	6.5 mm
Diameter of standard mounting	5 mm	5mm	10mm
Positioning stroke	5.7 mm	15 mm	10 mm
Fine positioning stroke	10 nm/V	20 nm/V	20 nm/V
Resolution	atomic	atomic	atomic
Repeatability per direction (at constant load)	about 1%	about 1%	about 1%
Driving force	0.03 N @3 g	0.03 N @3 g	0.25 N @25 g
Maximum load orthogonal to positioning direction	0.5 N @50 g	0.5 N @50 g	1.0 N @100 g
Free axial hole, minimum diameter	0.35 mm	0.35 mm	1.3 mm
Max. positioning speed	2 mm/s	5 mm/s	5 mm/s
Mass including standard mounting	2.20 g	2.63 g	9.77 g
Min. operation voltage, bipolar & symmetric	+/-15 V	+/-15 V	+/-15 V

<http://nanomotor.de/nanomotor>

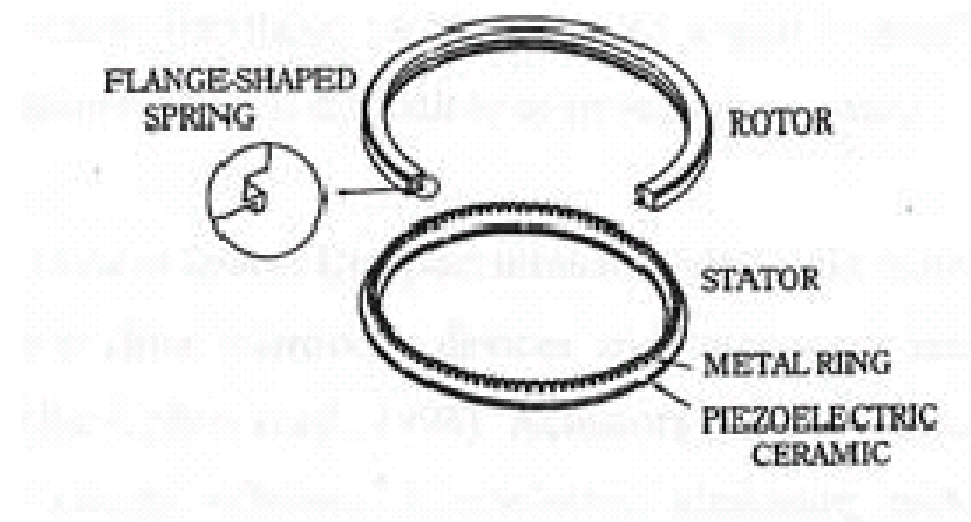
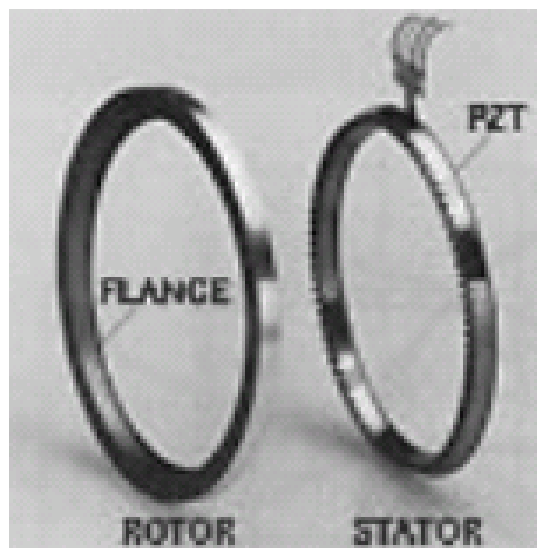
Nanomotor_brochure



Application

Auto focus camera systems

- Lighter than a similar electromagnetic system.
- Able to directly drive the focus rather than having a gear system.
- First practical use of ultrasonic motors.

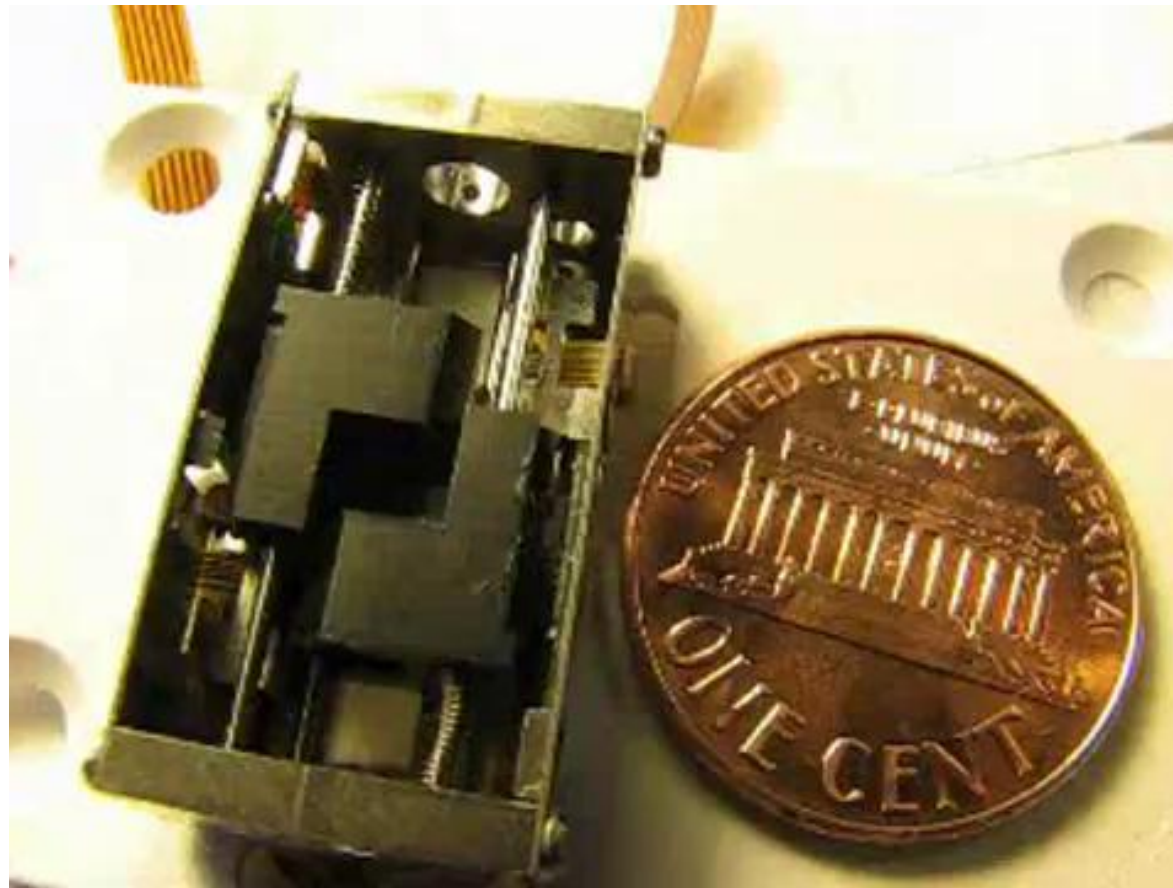


<http://www.photoscene.com/sw/tour/inside.htm>



Application

Optical zoom module



Application



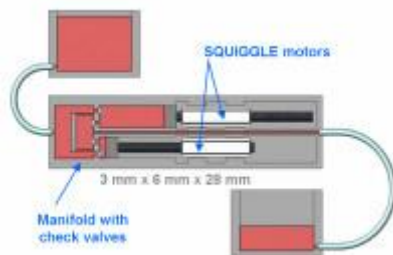
Development and construction of a mobile robot driven by miniature ultrasonic piezoelectric motors for **minimally invasive cardiac therapy**. The robot design extends upon previous prototypes of HeartLander miniature mobile robot that moves in an inchworm-like fashion (http://www.newscaletech.com/article_papers.html#HeartLanderPaper)



The SQUIGGLE motor is **non-magnetic** and can be used near MRI systems. In this setup the screw of the motor pushes a syringe plunger, to deliver a contrast media during a scan. (<https://newscaletech.com/resources/technology/squiggle-micro-motor-technology/>)



Ultra-thin **autofocus lens** module prototype
http://www.newscaletech.com/phone_camera.html



<http://www.newscaletech.com/microfluidics.html>



References and further readings

- Pons, J.L., “Emerging Actuator Technologies: A Micromechatronic Approach”, John Wiley & Sons, Ltd, England, 2005
- M. Karpelson, et al., Driving high voltage piezoelectric actuators in microrobotic applications, Sens. Actuators A: Phys. (2012)

